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# Supplementary theory and analyses for “Scale-dependent contraction of spatial wet-bulb temperature contrasts in eastern China”

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This supplement provides the proof of Proposition 1, a process-level strong-mixing result, additional applied analyses and the simulation specifications and results. Unless stated otherwise, evaluation-period tables use 33 summers and exclude the 2015 and 2022 method-development summers. The historical and station extensions state their own temporal supports.

## Analysis sequence

Table S1 summarises the sequence of analyses. Unless stated otherwise, each calculation uses the 121-site labels, five bandwidths and equal-month, equal-scale and equal-summer estimator. The historical and station extensions were designed before their respective data were retrieved. The protocol-defined status of an analysis is retained below even when later diagnostics changed how it is interpreted in the final paper.

The archived protocol file states that Version 2 was frozen on 2 August 2026, before acquisition or inspection of the 33 evaluation summers. Its SHA-256 is [13c3a32532921e9230d00e88e07bb1aaa95e1a27ac970d843739bbb3090a3a70](#). The first commit visible in the public Git history is later, on 11 August 2026 (2464330). The public Git history therefore does not independently establish the file's stated freeze date. The submission release at <https://github.com/Stork343/eastern-china-wet-bulb-contraction/releases/tag/jrssc-submission-v1-2026-08-13> archives the protocol, its digest, this status table and the submitted analysis together.

The surviving project record identifies 2015 and 2022 as the only complete summers used to develop and correct the method, but it preserves no contemporaneous rationale for why those two summers were acquired first. The wet-bulb fields and graph profiles from both summers were inspected during development. They were then excluded from the 33-summer evaluation period; no claim that they were random or representative development summers is made.

Table S1: Protocol status, timing, observed result and final interpretive role.

| Element                                                                                                           | Original status                 | Fixed before the 33-summer results?          | Observed result and final role                                        | Reason or scope                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------|---------------------------------|----------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Field construction, event labels, five graph scales and aggregation                                               | Primary specification           | Yes                                          | Retained; defines the finite-record target                            | Developed using 2015 and 2022, which were excluded from evaluation                                                    |
| 33-summer evaluation-period contrast                                                                              | Primary estimand                | Yes                                          | −7.2834%; primary finite-record summary                               | Equal months, scales and summers                                                                                      |
| Ordinary Student mean test                                                                                        | Confirmatory component          | Yes                                          | One-sided $p = 2.56 \times 10^{-6}$ ; process reference               | Later simulations showed undercoverage under appreciable interannual dependence                                       |
| Lag-2 Newey–West test                                                                                             | Confirmatory component          | Yes                                          | One-sided $p = 3.54 \times 10^{-6}$ ; process reference               | A fixed lag does not by itself ensure consistent long-run variance estimation                                         |
| Independent-summer exact binomial sign reference                                                                  | Confirmatory component          | Yes                                          | 28/33 negative; $p = 3.31 \times 10^{-5}$ ; recurrence reference      | Its exact-binomial interpretation requires independent summer signs                                                   |
| Three-component consistency rule                                                                                  | Confirmatory rule               | Yes                                          | confirmatory_consistency = TRUE at the prespecified 0.025 threshold   | All three protocol components passed; reported transparently but not treated as definitive long-run process inference |
| Global 99,999-draw product shift                                                                                  | Not in the protocol             | No; added after observing the primary result | Exploratory conditional randomisation; $p_{MC} = 10^{-5}$             | Applies only under the conditional product-invariance null                                                            |
| Process-model references, alternative effect measures, spatial bases, bandwidth and support comparisons 1950–1990 | Exploratory analyses            | No, except where separately dated            | Reference inference, structural descriptions and sensitivity analyses | Do not replace the protocol-defined raw-ratio target                                                                  |
| ERA5-Land extension                                                                                               | Pre-access extension            | Before access to historical values           | Retained as a temporally separated extension                          | Shares the ERA5-Land system with the evaluation record                                                                |
| Historical climatology–anomaly decomposition                                                                      | Exploratory structural analysis | No                                           | Period-specific algebraic accounting                                  | Descriptive, not a physical-mechanism analysis                                                                        |
| Ten-year NOAA–ERA5-Land extension                                                                                 | Pre-access extension            | Before station results were retrieved        | Sparse external measurement comparison                                | ERA5-Land defines events; not an independent event replication                                                        |
| Heavy-tail, coupled-field and cross-record simulations                                                            | Exploratory simulations         | After method development                     | Estimator behaviour and conditional-randomisation calibration         | A simulated departure cannot verify product invariance in the observed climate record                                 |

## S1. Finite-record product cyclic-shift validity

Let  $X$  collect the regional-mean records used to define the event labels, and let  $D$  collect the corresponding daily graph-dispersion profiles. The finite group  $G$  acts on  $D$  by independently rotating the complete multiscale profile within each month-year record. Conditional on  $X$ , suppose

$$(D \mid X) \stackrel{d}{=} (gD \mid X) \quad \text{for every } g \in G. \quad (\text{S1})$$

In nontechnical terms, conditional on the observed regional-mean series, independently changing the circular phase of the complete dispersion profile in any month-year record must leave the joint distribution unchanged. This is a conditional phase-invariance null. It is stronger than  $E(Q^H - Q^M) = 0$  and is not a general null of no association between regional mean and spatial dispersion. Common weather modes across months, circulation or soil-moisture persistence, and other shared seasonal or interannual drivers can violate the product condition because independent record-wise rotations break their phase alignment.

For a measurable lower-tail statistic  $T(X, D)$  fixed before the transformations are examined, define

$$p_G(X, D) = \frac{1}{|G|} \sum_{g \in G} I\{T(X, gD) \leq T(X, D)\}.$$

**Proposition 1** (Product cyclic-shift validity) *For every  $u \in [0, 1]$ ,*

$$\Pr\{p_G(X, D) \leq u \mid X\} \leq u \quad \text{almost surely.}$$

*If the  $|G|$  transformed statistics have no ties,  $p_G$  is uniform on  $\{1/|G|, \dots, 1\}$  conditional on the invariant orbit. If  $H_1, \dots, H_B$  are independent uniforms on  $G$ , independently of  $(X, D)$ , then*

$$p_B = \frac{1 + \sum_{b=1}^B I\{T(X, H_b D) \leq T(X, D)\}}{B + 1}$$

*satisfies*

$$\Pr\{p_B \leq u \mid X\} \leq \frac{\lfloor (B+1)u \rfloor}{B+1} \leq u.$$

*For fixed scale statistics  $T_1, \dots, T_H$ , put  $M(X, D) = \min_h T_h(X, D)$  and define*

$$p_h^{\text{adj}} = \frac{1}{|G|} \sum_{g \in G} I\{M(X, gD) \leq T_h(X, D)\}.$$

*Under the complete joint invariance null,  $\Pr\{\min_h p_h^{\text{adj}} \leq u \mid X\} \leq u$ . This result provides weak family-wise control for the complete joint null.*

*Proof* Fix a value  $x$  in the probability-one set on which the conditional invariance in Equation (S1) holds, and write  $S(y) = T(x, y)$ . Let  $m = |G|$ .

For fixed  $y$ , form the multiset  $\{S(gy) : g \in G\}$ . For each  $g \in G$ , let

$$r_g(y) = \sum_{h \in G} I\{S(hy) \leq S(gy)\}.$$

Right multiplication by  $g$  permutes the elements of  $G$ . Hence

$$\begin{aligned} p_G(x, gy) &= \frac{1}{m} \sum_{h \in G} I\{S(hgy) \leq S(gy)\} \\ &= \frac{r_g(y)}{m}. \end{aligned}$$

If  $U$  is uniform on  $G$ , independently of  $y$ , then

$$\Pr\{p_G(x, Uy) \leq u \mid y\} = \frac{1}{m} \#\{g : r_g(y) \leq mu\}. \quad (\text{S2})$$

Order the  $m$  values in the multiset. An observation at the upper end of a tie block has weak rank equal to the final position of that block. Every tie block counted on the right of Equation (S2) therefore ends no later than position  $\lfloor mu \rfloor$ . The number of counted group elements is at most  $\lfloor mu \rfloor$ , and

$$\Pr\{p_G(x, Uy) \leq u \mid y\} \leq \frac{\lfloor mu \rfloor}{m} \leq u.$$

This inequality also holds when the action has a nontrivial stabiliser. Repeated transformations then appear as ties in the multiset and make the weak-rank test conservative.

Let  $U$  now be independent of  $(X, D)$  and uniform on  $G$ . Conditional invariance gives  $UD \mid X = x \stackrel{d}{=} D \mid X = x$ . Applying the preceding inequality conditionally on  $D$  yields

$$\begin{aligned} \Pr\{p_G(x, D) \leq u \mid X = x\} &= \Pr\{p_G(x, UD) \leq u \mid X = x\} \\ &= \mathbb{E}[\Pr\{p_G(x, UD) \leq u \mid X = x, D\} \mid X = x] \\ &\leq u. \end{aligned}$$

If the transformed statistics have no ties, the ranks  $r_g(y)$  are  $1, \dots, m$  once each. Equality holds at  $u = j/m$ , proving uniformity on the attainable grid.

For the sampled version, take  $U_0, U_1, \dots, U_B$  independently and uniformly from  $G$ , independently of  $(X, D)$ . Conditional on  $(X, D)$ , the variables  $S(U_0D), \dots, S(U_BD)$  are identically distributed and exchangeable. The weak rank of the first among these  $B + 1$  values satisfies

$$\Pr \left[ \frac{1 + \sum_{b=1}^B I\{S(U_bD) \leq S(U_0D)\}}{B + 1} \leq u \mid X = x \right] \leq \frac{\lfloor (B + 1)u \rfloor}{B + 1}. \quad (\text{S3})$$

It remains to relate the random anchor to the implemented construction, whose first transformation is the identity. Put  $H_b = U_b U_0^{-1}$  for  $b = 1, \dots, B$  and  $D^* = U_0 D$ . The map

$$(U_0, U_1, \dots, U_B) \mapsto (U_0, H_1, \dots, H_B)$$

is a bijection on  $G^{B+1}$ . Thus  $H_1, \dots, H_B$  are independent uniforms and are independent of  $U_0$ . Moreover,  $D^* \mid X = x \stackrel{d}{=} D \mid X = x$ . The vector in Equation (S3) consequently has the same conditional distribution as

$$\{S(D), S(H_1D), \dots, S(H_BD)\}.$$

This is the identity-plus- $B$  construction used to calculate  $p_B$ .

The statistic must remain a fixed measurable function of the jointly shifted outcome matrix. For the scale-wise single-step comparison, let  $T_h$  denote the scale statistics and put  $M(y) = \min_h T_h(x, y)$ . The adjusted value

$$p_h^{\text{adj}} = \frac{1}{m} \sum_{g \in G} I\{M(gy) \leq T_h(x, y)\}$$

is nondecreasing in its observed threshold. Therefore  $\min_h p_h^{\text{adj}}$  is attained at a scale for which  $T_h(x, y) = M(y)$  and equals the randomisation value for  $M$ . Applying the enumeration result to  $M$  gives

$$\Pr\{\min_h p_h^{\text{adj}} \leq u \mid X\} \leq u.$$

The conclusion uses the complete joint null because  $M$  combines every scale. This completes the proof.  $\square$

## S2. Process-level strong-mixing and HAC reference

Let  $\hat{\mathbf{R}}_r = (\hat{R}_{h_1r}, \dots, \hat{R}_{h_Hr})^\top$  be the scale profile for summer  $r$ ,  $\mathbf{a} = H^{-1}\mathbf{1}_H$ ,  $Z_r = \mathbf{a}^\top \hat{\mathbf{R}}_r$  and  $\hat{\theta}_R = R^{-1} \sum_{r=1}^R Z_r$ . Write  $\gamma(k) = \text{Cov}(Z_0, Z_k)$  and define

$$\begin{aligned} \hat{\gamma}_R(k) &= \frac{1}{R} \sum_{r=k+1}^R (Z_r - \bar{Z}_R)(Z_{r-k} - \bar{Z}_R), \\ \hat{\sigma}_R^2 &= \hat{\gamma}_R(0) + 2 \sum_{k=1}^{L_R} K\left(\frac{k}{L_R + 1}\right) \hat{\gamma}_R(k). \end{aligned} \quad (\text{S4})$$

**Theorem S1** (Strong-mixing limit and HAC reference) Suppose  $\{\widehat{\mathbf{R}}_r\}_{r \in \mathbb{Z}}$  is strictly stationary and strongly mixing, with  $E\|\widehat{\mathbf{R}}_0\|^{2+\delta} < \infty$  for some  $\delta > 0$  and  $\sum_{k=1}^{\infty} \alpha(k)^{\delta/(2+\delta)} < \infty$ . Then every entry of

$$\mathbf{\Omega} = \sum_{k=-\infty}^{\infty} \text{Cov}(\widehat{\mathbf{R}}_0, \widehat{\mathbf{R}}_k)$$

is absolutely convergent. If  $\sigma^2 = \mathbf{a}^\top \mathbf{\Omega} \mathbf{a} > 0$  and  $\theta = \mathbf{a}^\top E(\widehat{\mathbf{R}}_0)$ , then

$$\sqrt{R}(\widehat{\theta}_R - \theta) \xrightarrow{d} N(0, \sigma^2).$$

Suppose in addition that  $K$  is bounded on  $[0, \infty)$ ,  $K(0) = 1$ , is continuous at zero and vanishes outside  $[0, 1]$ ;  $L_R \rightarrow \infty$ ,  $L_R/R \rightarrow 0$ ; and

$$\sum_{k=0}^{L_R} |\widehat{\gamma}_R(k) - \gamma(k)| = o_p(1). \quad (\text{S5})$$

Then  $\widehat{\sigma}_R^2 \xrightarrow{p} \sigma^2$  and

$$\frac{\sqrt{R}(\widehat{\theta}_R - \theta)}{\widehat{\sigma}_R} \xrightarrow{d} N(0, 1).$$

For a ratio component  $\widehat{R}_{jr} = A_{jr}/B_{jr} - 1$ , with  $A_{jr} \geq 0$  and  $B_{jr} > 0$ , a sufficient condition for its  $(2 + \delta)$  moment is the existence of conjugate  $p, q > 1$  such that

$$E A_{j0}^{(2+\delta)p} < \infty, \quad E B_{j0}^{-(2+\delta)q} < \infty.$$

The same conclusion holds for a finite average of such ratios when the condition holds for every constituent ratio.

*Proof* Let  $\boldsymbol{\mu} = E(\widehat{\mathbf{R}}_0)$ . A measurable transformation cannot increase strong-mixing coefficients, so  $\{Z_r\}$  is strictly stationary and strongly mixing with coefficients bounded by  $\alpha(k)$ . The inequality  $|\mathbf{a}^\top v| \leq \|\mathbf{a}\| \|v\|$  gives  $E|Z_0|^{2+\delta} < \infty$ .

Put  $\xi_r = Z_r - E Z_0$ . Davydov's covariance inequality, with both moment exponents equal to  $2 + \delta$ , gives a finite constant  $C$  such that

$$|\gamma(k)| = |E(\xi_0 \xi_k)| \leq C \|\xi_0\|_{2+\delta}^2 \alpha(|k|)^{\delta/(2+\delta)}, \quad k \neq 0. \quad (\text{S6})$$

The mixing-rate assumption implies  $\sum_{k=-\infty}^{\infty} |\gamma(k)| < \infty$ . Applying the same inequality componentwise establishes absolute convergence of every entry of  $\mathbf{\Omega}$ . Bilinearity and absolute convergence permit interchange of the finite quadratic form and the infinite sum:

$$\mathbf{a}^\top \mathbf{\Omega} \mathbf{a} = \sum_{k=-\infty}^{\infty} \text{Cov}(Z_0, Z_k) = \sum_{k=-\infty}^{\infty} \gamma(k) = \sigma^2. \quad (\text{S7})$$

The central limit theorem for strongly mixing sequences applies under the stated moment and mixing-rate conditions (Ibragimov and Linnik, 1971). Hence

$$R^{-1/2} \sum_{r=1}^R \xi_r \xrightarrow{d} N(0, \sigma^2).$$

Since  $E Z_0 = \mathbf{a}^\top \boldsymbol{\mu} = \theta$  and  $R^{-1} \sum_r Z_r = \widehat{\theta}_R$ , this proves the first limit.

For consistency of Equation (S4), define

$$\sigma_{K,L_R}^2 = \gamma(0) + 2 \sum_{k=1}^{L_R} K \left( \frac{k}{L_R + 1} \right) \gamma(k).$$

Boundedness of  $K$  and Equation (S5) give

$$\begin{aligned} |\hat{\sigma}_R^2 - \sigma_{K,L_R}^2| &\leq |\hat{\gamma}_R(0) - \gamma(0)| \\ &+ 2\|K\|_\infty \sum_{k=1}^{L_R} |\hat{\gamma}_R(k) - \gamma(k)| = o_p(1). \end{aligned}$$

For each fixed  $k$ , continuity at zero and  $L_R \rightarrow \infty$  imply  $K\{k/(L_R + 1)\} \rightarrow 1$ . The weights are uniformly bounded, vanish for  $k > L_R$ , and  $\sum_k |\gamma(k)| < \infty$ . Dominated convergence for series therefore yields

$$\sigma_{K,L_R}^2 \longrightarrow \gamma(0) + 2 \sum_{k=1}^{\infty} \gamma(k) = \sigma^2.$$

Thus  $\hat{\sigma}_R^2 \xrightarrow{p} \sigma^2$ . Positivity of  $\sigma^2$  and Slutsky's theorem give the studentised limit.

Condition (S5) is separate from the assumptions used for the central limit theorem. Primitive sufficient conditions for it require higher moments, stronger mixing-rate summability and a compatible bandwidth rate (Andrews, 1991; Newey and West, 1987).

For comparison, fix a lag  $L$ . Strong mixing implies ergodicity. The ergodic theorem applied to  $(Z_r - E Z_0)(Z_{r-k} - E Z_0)$ , together with  $\bar{Z}_R \xrightarrow{p} E Z_0$ , gives  $\hat{\gamma}_R(k) \xrightarrow{p} \gamma(k)$  for each fixed  $k$ . The fixed-lag estimator therefore converges to

$$\sigma_{K,L}^2 = \gamma(0) + 2 \sum_{k=1}^L K \left( \frac{k}{L+1} \right) \gamma(k). \quad (\text{S8})$$

For Bartlett lag  $L = 2$ , this is  $\gamma(0) + 2\{(2/3)\gamma(1) + (1/3)\gamma(2)\}$ . Its difference from the full long-run variance is

$$\sigma^2 - \sigma_{K,2}^2 = 2 \left\{ \frac{1}{3}\gamma(1) + \frac{2}{3}\gamma(2) + \sum_{k=3}^{\infty} \gamma(k) \right\}.$$

The fixed-lag estimator is consistent only if this weighted error is zero.

It remains to verify the stated sufficient ratio condition. With  $s = 2 + \delta$ , convexity and Hölder's inequality give

$$\begin{aligned} E |A_{j0}/B_{j0} - 1|^s &\leq 2^{s-1} \left\{ E(A_{j0}^s B_{j0}^{-s}) + 1 \right\} \\ &\leq 2^{s-1} \left[ \{E A_{j0}^{sp}\}^{1/p} \{E B_{j0}^{-sq}\}^{1/q} + 1 \right] < \infty. \end{aligned}$$

A finite sum of variables with finite sth moments also has a finite sth moment. Applying this argument to each component proves the final claim.  $\square$

The 495 observed middle-day denominators range from 2.25 to 25.99 °C<sup>2</sup>. The process-level ratio argument additionally assumes the negative moment stated in Theorem S1.

### S3. Global product-shift result

The product-shift analysis used the statistic  $T = \hat{\theta}_{y_H}$ . Each draw independently selected one cyclic offset in each of the 99 evaluation month-year records and shifted all five graph-dispersion columns together. Regional-mean series

and event labels remained in their observed order. The statistic was then reconstructed with equal months, scales and summers. The generator seed was 20260809 and  $B = 99,999$ . This analysis was added after the primary 33-summer result was observed and is reported as exploratory.

No sampled statistic was at or below the observed  $-7.2834\%$ ; the lower-tail exceedance count was zero. The plus-one Monte Carlo value is therefore  $p_{MC} = (0 + 1)/(99,999 + 1) = 1.0 \times 10^{-5}$ , the minimum attainable value with 99,999 draws. The sampled null had mean  $0.8241\%$ , standard deviation  $1.5651$  points and median  $0.8139\%$ ; its  $0.1\%$ ,  $1\%$  and  $5\%$  quantiles were  $-3.8788\%$ ,  $-2.7440\%$  and  $-1.7225\%$ . A ratio-based randomisation distribution has its own empirical centre, and the test compares the observed statistic directly with that distribution.

**Table S2.** The 99,999-draw product cyclic-shift analysis. Effects, null means and null standard deviations are in percentage points. The last column uses the minimum statistic over the five scales and reports weak family-wise validity under the complete joint invariance null.

| Bandwidth<br>(km) | Observed<br>(points) | Null mean<br>(points) | Null SD<br>(points) | Unadjusted $p$ | Complete-null $p$ |
|-------------------|----------------------|-----------------------|---------------------|----------------|-------------------|
| 126               | -2.856               | 0.359                 | 1.059               | 0.00104        | 0.02504           |
| 252               | -3.945               | 0.531                 | 1.301               | 0.00022        | 0.00571           |
| 503               | -5.907               | 0.859                 | 1.683               | 0.00002        | 0.00019           |
| 1,006             | -10.441              | 1.125                 | 1.964               | 0.00001        | 0.00001           |
| 2,013             | -13.268              | 1.246                 | 2.089               | 0.00001        | 0.00001           |
| Five-scale mean   | -7.283               | 0.824                 | 1.565               | 0.00001        | –                 |

The relevant null is the conditional product-invariance statement in Equation (S1). The value is not evidence against every zero-mean or no-association null and does not establish a causal or universal climate-process conclusion. The scale-wise rows use the same joint draws and are mutually dependent. The cross-record stress experiment in Section S11.4 assesses one specified shared-process departure; near-nominal rejection in that experiment cannot establish that the observed records satisfy product invariance.

## S4. Physical scale of the graph contrast

For a field  $y$ ,

$$Q_h(y) = \frac{\sum_{i < j} w_{ij,h} (y_i - y_j)^2}{2 \sum_{i < j} w_{ij,h}}, \quad D_h^{\text{RMS}}(y) = \{2Q_h(y)\}^{1/2}.$$

Thus  $Q_h$  is one half of the edge-weighted mean squared pairwise difference, equivalently an edge-weighted semivariance. It has units  $^{\circ}\text{C}^2$ , whereas  $D_h^{\text{RMS}}$  is the weighted root mean squared pairwise difference in  $^{\circ}\text{C}$ . Table S3 reports equal-month and equal-summer means of  $Q_h$  and applies  $D_h^{\text{RMS}} = (2Q_h)^{1/2}$  to each displayed mean. The relative columns average record-level ratios and therefore differ from ratios of the displayed marginal means.

**Table S3.** Physical scale of graph dispersion in the evaluation summers. The RMS columns are  $(2Q_h)^{1/2}$  transformations of the displayed mean  $Q_h$  columns. The interval is a 95%  $t$ -based interval that treats summers as independent. RMS change is the mean record-level relative change in  $D_h^{\text{RMS}}$ .

| Bandwidth<br>(km) | $Q_h^M$<br>( $^{\circ}\text{C}^2$ ) | $Q_h^H$<br>( $^{\circ}\text{C}^2$ ) | $\text{RMS}^M$<br>( $^{\circ}\text{C}$ ) | $\text{RMS}^H$<br>( $^{\circ}\text{C}$ ) | $Q_h$ effect [interval]<br>(%) | RMS change<br>(%) |
|-------------------|-------------------------------------|-------------------------------------|------------------------------------------|------------------------------------------|--------------------------------|-------------------|
| 126               | 2.662                               | 2.578                               | 2.307                                    | 2.271                                    | -2.86 [-5.04, -0.67]           | -1.54             |
| 252               | 4.356                               | 4.169                               | 2.952                                    | 2.887                                    | -3.94 [-6.50, -1.39]           | -2.14             |
| 503               | 7.943                               | 7.433                               | 3.986                                    | 3.856                                    | -5.91 [-8.91, -2.90]           | -3.25             |
| 1,006             | 12.892                              | 11.489                              | 5.078                                    | 4.793                                    | -10.44 [-13.49, -7.39]         | -5.70             |
| 2,013             | 15.903                              | 13.743                              | 5.640                                    | 5.243                                    | -13.27 [-16.28, -10.26]        | -7.22             |

The direct five-scale mean of the record-level RMS changes is  $-3.97\%$ . Applying the square-root transformation only after averaging the five graph-dispersion effects gives  $\sqrt{1 - 0.072834} - 1 = -3.71\%$ . The two values differ because

the square root and the averaging steps do not commute. In either form, a percentage change in squared pairwise differences is larger in magnitude than the corresponding change in RMS amplitude.

The five quoted distances are Gaussian bandwidths  $h$ , not observed pair distances or hard cut-offs. At  $h = 126, 252, 503, 1,006$  and  $2,013$  km, the edge-weighted mean pair distances are respectively 200, 319, 549, 826 and 992 km; the corresponding Kish effective edge counts are 366, 1,138, 3,223, 6,005 and 7,109 out of 7,260. The five-scale summary is an equal-weight mean over this prespecified logarithmic bandwidth grid. It is a defined summary of that grid, not a unique natural regional index.

## S5. Structural decompositions

### S5.1. Climatology–anomaly decomposition

Let  $\mu_m$  be the site-specific 35-summer mean field for month  $m$  and write  $y_t = \mu_m + a_t$ . For a fixed graph,

$$Q_h(y_t) = \frac{\mu_m^\top L_h \mu_m}{2S_h} + \frac{\mu_m^\top L_h a_t}{S_h} + \frac{a_t^\top L_h a_t}{2S_h}. \quad (S9)$$

The first term is fixed within month  $m$ , so its high-minus-middle difference is exactly zero. Within each record, we divide the differences of the other two terms by the middle-day total  $Q_h$  mean. The resulting anomaly-energy and climatology–anomaly alignment components add to the original relative contrast before any aggregation.

The middle term is a signed bilinear cross term and can be negative; it is not an energy. The decomposition localises the observed contrast algebraically with respect to a chosen climatology. It does not identify a circulation, soil-moisture or other physical mechanism.

**Table S4.** Exact climatology–anomaly decomposition. Entries are percentage points; brackets are 95% intervals. The climatology–energy difference is zero by construction.

| Bandwidth<br>(km) | Total effect<br>[interval] | Anomaly energy<br>[interval] | Climatology–anomaly alignment<br>[interval] |
|-------------------|----------------------------|------------------------------|---------------------------------------------|
| 126               | −2.86 [−5.04, −0.67]       | −2.63 [−3.55, −1.72]         | −0.22 [−1.98, 1.54]                         |
| 252               | −3.94 [−6.50, −1.39]       | −2.45 [−3.45, −1.45]         | −1.49 [−3.69, 0.70]                         |
| 503               | −5.91 [−8.91, −2.90]       | −1.42 [−2.43, −0.41]         | −4.49 [−7.24, −1.74]                        |
| 1,006             | −10.44 [−13.49, −7.39]     | −0.18 [−1.10, 0.74]          | −10.26 [−13.20, −7.32]                      |
| 2,013             | −13.27 [−16.28, −10.26]    | +0.41 [−0.50, 1.31]          | −13.68 [−16.68, −10.67]                     |

At 2,013 km, anomaly energy is lower in 16 of 33 summers, whereas the alignment component is negative in all 33. The corresponding counts at 1,006 km are 21 and 28. The broad-scale contraction is therefore associated with daily anomalies opposing the persistent monthly field, while the transient anomaly energy itself changes little. The decomposition is descriptive and conditional on the chosen climatology. The maximum aggregated identity error over the five graphs is  $1.82 \times 10^{-16}$ .

As a sensitivity analysis, the monthly climatology for each evaluated summer was recomputed from the other 34 summers. Table S5 compares this leave-one-summer-out construction with the inclusive 35-summer climatology. The broad-scale accounting is practically unchanged: at  $h = 2,013$  km the anomaly-energy component is +0.40 rather than +0.41 percentage points, and the signed cross component is −13.67 rather than −13.68 points.

### S5.2. Geographic and topographic bases

For a centred basis matrix  $B$ , define the graph-orthogonal projection

$$\hat{\beta}_{th} = (B^\top L_h B)^{-1} B^\top L_h y_t, \quad \hat{y}_{th} = B \hat{\beta}_{th}.$$



**Table S5.** Sensitivity of the exact decomposition to leave-one-summer-out monthly climatology. Entries are percentage points. “Cross” denotes the signed climatology–anomaly bilinear term, not an energy or mechanism.

| $h$ (km) | Inclusive anomaly | Inclusive cross | LOSO anomaly | LOSO cross |
|----------|-------------------|-----------------|--------------|------------|
| 126      | −2.63             | −0.22           | −2.65        | −0.20      |
| 252      | −2.45             | −1.49           | −2.47        | −1.47      |
| 503      | −1.42             | −4.49           | −1.44        | −4.47      |
| 1,006    | −0.18             | −10.26          | −0.19        | −10.25     |
| 2,013    | +0.41             | −13.68          | +0.40        | −13.67     |

The structured energy of  $\hat{\mathbf{y}}_{th}$  and the residual energy of  $\mathbf{y}_t - \hat{\mathbf{y}}_{th}$  add exactly to  $Q_h(\mathbf{y}_t)$ . Table S6 compares centred latitude, the latitude–longitude plane and a latitude–longitude–elevation basis. Elevation comes from the invariant ERA5-Land geopotential field: the nearest 0.1-degree grid value at each analysis site was divided by  $9.80665 \text{ m s}^{-2}$ . The 121 elevations range from  $-0.9$  to  $1,940.3$  m. Each basis column was centred and scaled before the projection. Such scaling leaves its column space unchanged.

**Table S6.** Nested spatial-basis comparison. Values are mean percentage-point contributions across the evaluation summers. The structured and residual columns for each basis add to the total, apart from rounding.

| Bandwidth<br>(km) | Total<br>effect | Latitude<br>structured | Latitude<br>residual | Planar<br>structured | Planar<br>residual | Geographic–<br>topographic<br>structured | Geographic–<br>topographic<br>residual |
|-------------------|-----------------|------------------------|----------------------|----------------------|--------------------|------------------------------------------|----------------------------------------|
| 126               | −2.86           | −2.58                  | −0.28                | −2.81                | −0.04              | −1.16                                    | −1.69                                  |
| 252               | −3.94           | −4.60                  | +0.66                | −5.07                | +1.13              | −2.85                                    | −1.10                                  |
| 503               | −5.91           | −8.38                  | +2.48                | −9.32                | +3.42              | −6.13                                    | +0.23                                  |
| 1,006             | −10.44          | −12.73                 | +2.29                | −13.94               | +3.50              | −11.39                                   | +0.94                                  |
| 2,013             | −13.27          | −14.96                 | +1.69                | −16.23               | +2.96              | −14.34                                   | +1.07                                  |

The latitude component is negative in 33 of 33 summers at every scale. The planar component is negative in 32 summers at 126 km and all 33 summers at the other four scales. At 2,013 km, its 95% interval is  $[-18.16, -14.30]$  points and the residual interval is  $[1.59, 4.33]$  points. Adding longitude makes the structured component more negative and the compensating residual more positive.

The geographic–topographic component is negative in 18, 21, 28, 33 and 33 of the 33 summers as bandwidth increases. At 126 km its estimate is  $-1.16$  points with a 95% interval of  $[-2.66, 0.33]$ , which includes zero. At 2,013 km the estimate is  $-14.34$  points  $[-16.75, -11.92]$ , and the residual is  $+1.07$  points. The broad contraction persists after adding elevation to the latitude–longitude basis. The projection yields a combined geographic–topographic contribution; no order-invariant elevation contribution or physical mechanism follows from it. The maximum day-level identity error for the augmented basis is  $8.88 \times 10^{-14}$ , and the maximum aggregated error is  $2.78 \times 10^{-16}$ .

## S6. Scale and support resolution

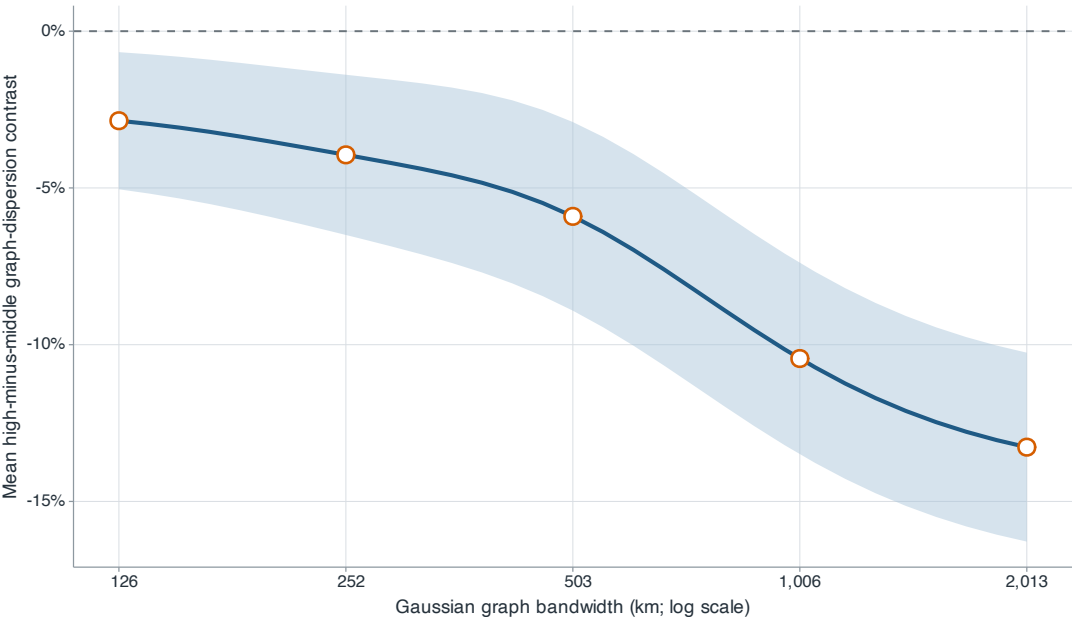
### S6.1. Thirty-one-point bandwidth curve

The estimand is the equal-weight mean over five design bandwidths. A denser curve evaluates 31 logarithmically spaced values between 125.800 and 2,012.796 km on the same 121 sites, using the evaluation fields and labels. The five design bandwidths define the estimand; the denser curve describes its scale profile.

All 30 adjacent changes are negative, and all 31 pointwise 95% intervals have upper endpoints below zero. The five design estimates remain  $-2.86, -3.94, -5.91, -10.44$  and  $-13.27\%$ . The curve shows no reversal at the 31 evaluated bandwidths. The intervals are descriptive pointwise summaries; no simultaneous confidence band is estimated.

**Table S7.** The 31-point bandwidth curve. Values are mean graph-dispersion effects in per cent. The three column pairs continue the same ordered curve.

| <i>h</i> (km) | Effect | <i>h</i> (km) | Effect | <i>h</i> (km) | Effect  |
|---------------|--------|---------------|--------|---------------|---------|
| 125.8         | −2.856 | 347.7         | −4.596 | 876.1         | −9.538  |
| 138.0         | −2.958 | 381.4         | −4.835 | 961.0         | −10.151 |
| 151.3         | −3.076 | 418.3         | −5.125 | 1,054.0       | −10.720 |
| 166.0         | −3.211 | 458.8         | −5.479 | 1,156.0       | −11.239 |
| 182.1         | −3.360 | 503.2         | −5.907 | 1,268.0       | −11.703 |
| 199.7         | −3.521 | 551.9         | −6.407 | 1,390.8       | −12.113 |
| 219.0         | −3.688 | 605.4         | −6.974 | 1,525.4       | −12.470 |
| 240.2         | −3.859 | 664.0         | −7.591 | 1,673.1       | −12.778 |
| 263.5         | −4.031 | 728.3         | −8.239 | 1,835.1       | −13.043 |
| 289.0         | −4.206 | 798.8         | −8.895 | 2,012.8       | −13.268 |
| 317.0         | −4.391 |               |        |               |         |



**Figure 1** Bandwidth-resolution profile on the 121-site support. The blue line and shaded band show the 31-point profile and its pointwise 95% intervals; orange open circles mark the five design bandwidths.

S6.2. Four spatial-support configurations

The convergence comparison uses the five physical bandwidths, evaluation peak times and 121-site labels. The 239-site configuration refines longitude; the 236-site configuration refines latitude. Each contains all 121 sites and is a subset of the 465-site grid. The two one-direction refinements differ in their added sites.

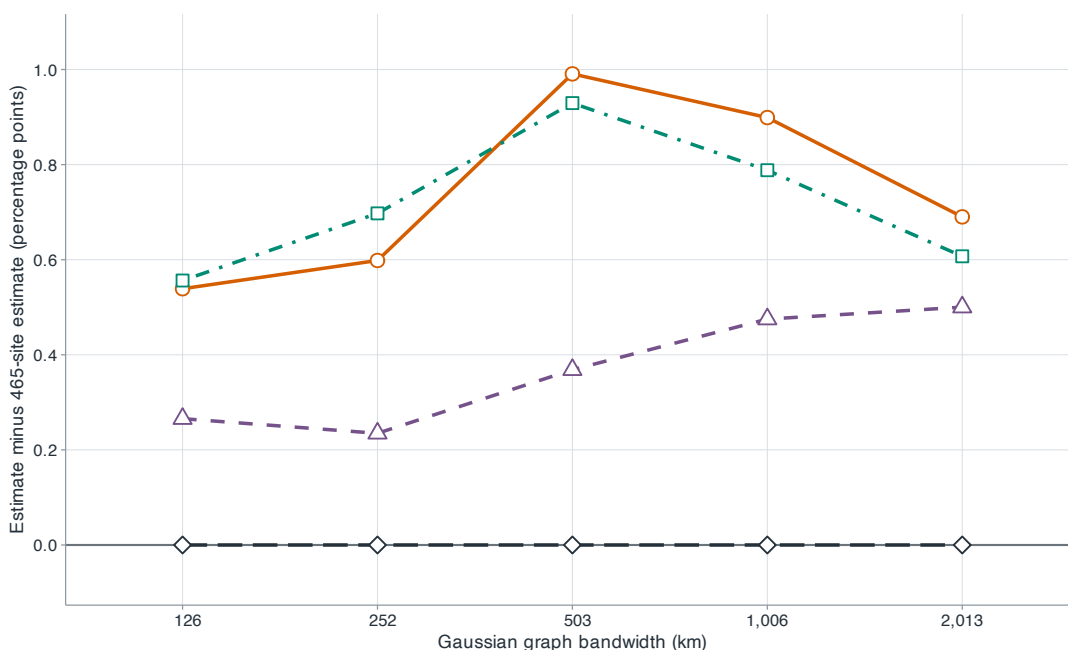
**Table S8.** Four-configuration spatial-resolution comparison. Effects and the two error summaries are in percentage points. Differences use the 465-site result at the same physical bandwidth as the reference.

| Configuration     | Sites | 126    | 252    | 503    | 1,006   | 2,013   | Max.  Δ | RMSE  |
|-------------------|-------|--------|--------|--------|---------|---------|---------|-------|
| Analysis support  | 121   | −2.856 | −3.945 | −5.907 | −10.441 | −13.268 | 0.991   | 0.764 |
| Longitude refined | 239   | −2.839 | −3.846 | −5.968 | −10.552 | −13.351 | 0.929   | 0.728 |
| Latitude refined  | 236   | −3.129 | −4.308 | −6.529 | −10.865 | −13.458 | 0.500   | 0.384 |
| Dense support     | 465   | −3.395 | −4.543 | −6.898 | −11.340 | −13.959 | 0.000   | 0.000 |

Every support retains the broad-to-local ordering and a negative effect at all five bandwidths. Relative to 465 sites, the 121-site profile has a maximum absolute discrepancy of 0.99 points and an across-scale RMSE of 0.76 points.

These finite-grid comparisons show how the estimates change with spatial support. Agreement is weakest for the narrowest graph, consistent with its shorter length scale relative to intersite spacing.

These convergence calculations use the primary equirectangular distance matrix. On the 465-site grid, the original evaluation events give  $-8.03\%$ ; reselecting events with the same distance matrix gives  $-7.25\%$ . The WGS84 relabelled calculation for the full 465-site domain, reported below, instead gives  $-6.93\%$ .



**Figure 2** Differences from the 465-site estimate at five bandwidths. Orange circles with a solid line denote the 121-site support; purple triangles with a dashed line, the latitude-refined 236-site support; teal squares with a dot-dash line, the longitude-refined 239-site support; and dark diamonds with a long-dash line, the dense 465-site reference.

### S6.3. Distance, boundary and non-rectangular support checks

Table S9 reports the complete five-bandwidth profiles from the additional support calculations. The 121-site WGS84 rows change the distance calculation and, where stated, the target weights and event labels. The remaining rows use the 465-site downloaded lattice, WGS84 distances, equal site weights and labels recomputed from each retained domain. For compactness, the table reports the larger of the two inward movements at each boundary; the retained results also contain both movements, fixed-label variants, cosine-latitude variants, weighted pair distances, effective edge counts and between-summer summaries.

For the 121-site cosine-latitude target, the primary equirectangular matrix gives  $-8.11\%$  with original labels and  $-7.67\%$  after relabelling. The corresponding WGS84 values in Table S9 are  $-7.80\%$  and  $-7.37\%$ .

All displayed profiles are negative and strengthen with bandwidth. Relabelled boundary means range from  $-6.32\%$  to  $-7.97\%$ ; the north boundary has the largest influence on magnitude. The Natural Earth row operationalises a non-rectangular target as the low-resolution feature named China, excluding the separately stored Taiwan feature, intersected with the downloaded lattice and its ERA5-Land-valid mask. These checks support the direction and scale ordering, but they also show that the magnitude belongs to the specified domain rather than to a unique definition of eastern China.

**Table S9.** Graph-dispersion contrasts under distance and spatial-support changes. Entries are percentages at the five Gaussian bandwidths; Mean is their equal-weight average. Boundary distances are inward movements because no fields were downloaded outside the original rectangle.

| Analysis                                              | Sites | 126   | 252   | 503   | 1,006  | 2,013  | Mean  |
|-------------------------------------------------------|-------|-------|-------|-------|--------|--------|-------|
| Primary equirectangular, equal-site, fixed labels     | 121   | -2.86 | -3.94 | -5.91 | -10.44 | -13.27 | -7.28 |
| WGS84, equal-site, fixed labels                       | 121   | -2.42 | -3.47 | -5.50 | -10.26 | -13.22 | -6.97 |
| WGS84, cosine-latitude, fixed labels                  | 121   | -3.11 | -4.33 | -6.45 | -11.14 | -14.00 | -7.80 |
| WGS84, cosine-latitude, relabelled                    | 121   | -2.95 | -4.04 | -6.00 | -10.54 | -13.31 | -7.37 |
| Full downloaded domain, relabelled                    | 465   | -2.44 | -3.34 | -5.61 | -10.26 | -13.02 | -6.93 |
| South edge inward 1.8°, relabelled                    | 457   | -2.28 | -2.99 | -5.11 | -9.62  | -12.22 | -6.44 |
| North edge inward 1.8°, relabelled                    | 413   | -2.57 | -3.91 | -7.00 | -11.94 | -14.41 | -7.97 |
| West edge inward 1.6°, relabelled                     | 415   | -1.97 | -2.31 | -4.55 | -9.85  | -12.90 | -6.32 |
| East edge inward 1.6°, relabelled                     | 458   | -2.44 | -3.22 | -5.42 | -10.07 | -12.80 | -6.79 |
| Natural Earth China-mainland intersection, relabelled | 445   | -2.09 | -2.80 | -5.06 | -9.64  | -12.18 | -6.35 |

## S7. Historical extension, 1950–1990

The ERA5-Land archive distributed through the Copernicus Climate Data Store begins in January 1950.<sup>1</sup> A plan dated 9 August 2026 defined the 41 summers, 121 sites, field construction, five scales, aggregation and product-shift calculation before the historical values were retrieved. The analysis weights the three months, five scales and summers equally. The two periods are separated in calendar time but share the ERA5-Land model and assimilation system. The product-shift calculation uses 99,999 draws, seed 20260810 and joint rotation of all five dispersion columns within each month-year record.

The five-scale contrast over 1950–1990 is -11.04% (descriptive  $t$ -reference interval [-12.84, -9.24]% based on between-summer variation), with a negative estimate in 40 of 41 summers. Deleting one summer at a time gives estimates from -11.33% to -10.62%. Table S10 gives the period split and the five-scale profile.

**Table S10.** The 1950–1990 historical extension. Estimates and intervals are percentages; the final column counts summers with a negative estimate. Period rows average the five bandwidths. Scale rows use all 41 summers.

| Period or bandwidth | Summers | Estimate | 95% interval     | Negative summers |
|---------------------|---------|----------|------------------|------------------|
| 1950–1990 overall   | 41      | -11.04   | [-12.84, -9.24]  | 40               |
| 1950–1978           | 29      | -10.50   | [-12.50, -8.51]  | 28               |
| 1979–1990           | 12      | -12.34   | [-16.65, -8.04]  | 12               |
| 126 km              | 41      | -5.96    | [-7.37, -4.54]   | 38               |
| 252 km              | 41      | -7.51    | [-9.15, -5.87]   | 39               |
| 503 km              | 41      | -10.08   | [-12.11, -8.04]  | 40               |
| 1,006 km            | 41      | -14.50   | [-16.72, -12.27] | 41               |
| 2,013 km            | 41      | -17.17   | [-19.44, -14.91] | 41               |

Provider documentation identifies the forcing change used for the period split: ERA5-Land for 1950–1978 used the preliminary ERA5 back extension, whereas the later period used the consolidated ERA5 release.<sup>2</sup> Both period estimates are negative, and their intervals overlap. The 1950–1978 estimate inherits uncertainty from the preliminary forcing data.

We applied the product cyclic-shift calculation to 123 month-year records. None of the sampled statistics was at or below the observed -11.043%, so the plus-one lower-tail value is 0.00001. The sampled null mean and standard deviation are 0.853% and 1.436 percentage points. Every scale has an unadjusted value of 0.00001; the complete-null single-step values range from 0.00001 to 0.00006.

<sup>1</sup> Dataset overview: <https://cds.climate.copernicus.eu/datasets/reanalysis-era5-land?tab=overview>.

<sup>2</sup> ERA5-Land production note: <https://confluence.ecmwf.int/pages/viewpage.action?pageId=402639006>.

For the historical climatology–anomaly decomposition, the monthly reference field was estimated within 1950–1990:

$$\mu_m^H = \frac{1}{41} \sum_{y=1950}^{1990} \left( \frac{1}{n_m} \sum_{t \in m} y_{yt} \right), \quad \mathbf{a}_{yt}^H = y_{yt} - \mu_m^H.$$

Thus  $\mu_m^H$  is the site-specific mean field for month  $m$  across the 41 historical summers. The anomaly and alignment terms therefore describe departures from the historical period's own persistent monthly spatial pattern.

Table S11 reports the latitude, planar and climatology–anomaly accounting for the historical extension. The latitude and planar structured components are negative in all 41 summers at every bandwidth. At 2,013 km, anomaly energy is  $-0.28$  points with interval  $[-1.01, 0.45]$ , while the climatology–anomaly alignment component is  $-16.89$  points  $[-19.31, -14.47]$  and is negative in all 41 summers. The broad-scale historical result is therefore also alignment-dominated, and its latitude and planar components have the same negative direction as in the 1991–2025 period.

**Table S11.** Structural accounting for the 1950–1990 extension. Entries are percentage points. Within each geographic basis, the structured and residual columns add to the total. The anomaly and alignment columns also add to the total because the climatology–energy difference is zero.

| Bandwidth<br>(km) | Total<br>effect | Latitude<br>structured | Latitude<br>residual | Planar<br>structured | Planar<br>residual | Anomaly<br>energy | Climatology–<br>anomaly alignment |
|-------------------|-----------------|------------------------|----------------------|----------------------|--------------------|-------------------|-----------------------------------|
| 126               | −5.96           | −2.83                  | −3.13                | −3.15                | −2.81              | −3.22             | −2.74                             |
| 252               | −7.51           | −4.92                  | −2.59                | −5.50                | −2.01              | −3.30             | −4.21                             |
| 503               | −10.08          | −8.86                  | −1.21                | −9.92                | −0.15              | −2.41             | −7.67                             |
| 1,006             | −14.50          | −13.60                 | −0.90                | −14.99               | +0.49              | −1.02             | −13.48                            |
| 2,013             | −17.17          | −16.13                 | −1.05                | −17.62               | +0.44              | −0.28             | −16.89                            |

## S8. NOAA station extension in ten evaluation years

The station comparison uses the ten summers enumerated before station retrieval: 1992, 1996, 2000, 2004, 2008, 2012, 2016, 2020, 2023 and 2025. The set is  $\{1992 + 4k : k = 0, \dots, 7\} \cup \{2023, 2025\}$ . The official Integrated Surface Database (ISD) station-history snapshot used for selection was archived.<sup>3</sup> The planned eligibility rule required a station-history end date on or after 31 August 2025, but that snapshot contained no such candidate: its global maximum end date was 28 August and its maximum inside the study rectangle was 24 August. The operational administrative cutoff was therefore 24 August. This administrative change preceded retrieval of station WBT observations and matched ERA5-Land values.

For a candidate station  $s$ , the proximity criterion was

$$d(s, \mathcal{S}_{121}) = \min_{i \in \mathcal{S}_{121}} d(s, i) \leq 150 \text{ km},$$

where  $\mathcal{S}_{121}$  is the 121-site analysis support. For coordinate differences in degrees, the pairwise distance was

$$d(s, i) = \left[ \{111.32 \cos(\bar{\phi}_{si})(\lambda_s - \lambda_i)\}^2 + \{110.57(\phi_s - \phi_i)\}^2 \right]^{1/2},$$

with  $\bar{\phi}_{si}$ , the pair's mean latitude, converted to radians in the cosine. This filter was applied before maximin selection and restricted station locations to within 150 km of the analysis support. Of the 228 stations satisfying the rectangle and date conditions, 20 exceeded 150 km, leaving 208 candidates; the largest retained nearest-site distance was 147.73 km and the smallest excluded distance was 155.72 km. A deterministic maximin rule based on station coordinates selected 30 stations. The study rectangle defined the candidate domain. A station-year qualified with

<sup>3</sup> NOAA Integrated Surface Database: <https://www.ncei.noaa.gov/products/land-based-station/integrated-surface-database>.

at least 400 exact UTC hours having temperature, dew point and usable station pressure; 26–30 of the 30 selected stations qualified in each year. Station wet-bulb temperature was calculated with the same Bolton implementation used for ERA5-Land. ISD sea-level pressure  $p_{SL}$  was converted to station pressure using the standard-atmosphere relation

$$p_s = 100p_{SL} \max\{1 - 2.25577 \times 10^{-5} z, 0.1\}^{5.2559},$$

where  $p_{SL}$  is in hPa and elevation  $z$  is in metres. Selected station elevations ranged from 3 to 2,063 m. Treating sea-level pressure as surface pressure instead changed WBT by 0.072 °C on average in the 175,172 matched hours; the maximum absolute difference was 1.346 °C.

The nearest ERA5-Land cells for the Shengsi and Taidong island stations had no finite land values at any requested time, so both point series were recorded as unavailable. The other 28 station locations had finite ERA5-Land series. Combining this land-mask restriction with the hourly qualification rule left 24–28 matched stations per year and 175,172 paired hours. Table S12 reports the measurement comparison. Bias is ERA5-Land minus station wet-bulb temperature; the centred correlation subtracts each station mean before pooling. The all-year row pools all matched hours across years.

**Table S12.** ERA5-Land and NOAA station wet-bulb agreement in ten evaluation years. Bias, MAE and RMSE are in °C. Stations are the qualified locations with a finite nearest-cell ERA5-Land series in that year.

| Year      | Stations | Matched hours | Bias  | MAE  | RMSE | Centred correlation |
|-----------|----------|---------------|-------|------|------|---------------------|
| 1992      | 25       | 17,887        | −0.25 | 0.88 | 1.16 | 0.919               |
| 1996      | 25       | 19,147        | −0.33 | 0.88 | 1.16 | 0.908               |
| 2000      | 25       | 16,765        | −0.40 | 0.94 | 1.21 | 0.908               |
| 2004      | 25       | 16,829        | −0.20 | 0.91 | 1.18 | 0.923               |
| 2008      | 25       | 17,310        | −0.20 | 0.88 | 1.14 | 0.909               |
| 2012      | 25       | 17,349        | −0.27 | 0.93 | 1.22 | 0.918               |
| 2016      | 24       | 16,589        | −0.34 | 0.88 | 1.13 | 0.937               |
| 2020      | 28       | 19,323        | −0.26 | 1.04 | 1.48 | 0.897               |
| 2023      | 27       | 16,554        | −0.19 | 1.02 | 1.49 | 0.925               |
| 2025      | 28       | 17,419        | −0.17 | 1.01 | 1.45 | 0.933               |
| All years | 28       | 175,172       | −0.26 | 0.94 | 1.27 | 0.916               |

For the effect comparison, ERA5-Land peak times and high/middle labels defined the event days, and we aligned station observations to those events. Each graph field used the same available station subset for NOAA and ERA5-Land and required at least ten stations. All 30 month-year records retained at least one high and one middle field, giving 345 usable event fields. We formed high-to-middle ratios within records, then averaged them equally over months within years and over the ten years. Table S13 reports 95% intervals. Inference focuses on the 503-, 1,006- and 2,013-km graphs supported by the station spacing; the two narrower graphs remain descriptive.

Availability was nearly the same by ERA5-Land-defined state: 122 of 240 scheduled high fields (50.8%) and 223 of 440 middle fields (50.7%) had at least ten stations. Their mean available-station counts over all scheduled fields were 13.28 and 13.12. Nevertheless, station support varied over time, and a retained record could have only one field in either state.

**Table S13.** High–middle graph-dispersion contrasts at ERA5-Land-defined peak times and labels in ten evaluation years. Contrasts and intervals are percentages. “Negative” counts years with a negative scale-specific estimate. Each scale contains all 30 month-year records and an average of 25.3 stations per event field.

| Bandwidth (km) | NOAA contrast [95% interval] | NOAA negative years | ERA5-Land contrast [95% interval] | ERA negative years |
|----------------|------------------------------|---------------------|-----------------------------------|--------------------|
| 126            | −6.23 [−25.87, 13.41]        | 6/10                | −11.60 [−27.49, 4.29]             | 7/10               |
| 252            | −7.23 [−21.01, 6.55]         | 7/10                | −12.74 [−22.68, −2.81]            | 8/10               |
| 503            | −12.50 [−24.05, −0.96]       | 8/10                | −17.13 [−25.15, −9.11]            | 10/10              |
| 1,006          | −18.03 [−29.17, −6.90]       | 9/10                | −21.60 [−29.07, −14.14]           | 10/10              |
| 2,013          | −21.47 [−32.51, −10.43]      | 9/10                | −24.23 [−31.72, −16.74]           | 10/10              |

The station and paired ERA5-Land profiles are negative at all five scales and become more negative with bandwidth. Averaging the three station-supported scales within each year gives  $-17.34\%$  (95% interval  $[-28.49, -6.18]\%$ ), with a negative average in nine of ten years. The same broad-profile calculation for the matched ERA5-Land fields gives  $-20.99\%$  ( $[-28.57, -13.40]\%$ ), negative in all ten years. The two fields are compared on identical, time-varying station supports. Because ERA5-Land defines the event times and labels, the comparison measures agreement for common events on a sparse station network. It is external measurement consistency, not an independent event definition or replication. Coverage is weakest at 126 and 252 km. Requiring two, three or five retained days per state gave broad-scale station summaries of  $-13.42\%$ ,  $-15.24\%$  and  $-14.56\%$ ; the last used only six years. A year-specific fixed common-station support gave  $-18.08\%$ . Defining daily peaks and high/middle labels from station regional means, with at least five days per state, gave  $-15.99\%$ . These variants retain the broad-scale negative direction but do not remove the limitations of the sparse, changing station network.

## S9. Continuous intensity and calendar matching

Two alternative event contrasts use the same sites, graphs and aggregation hierarchy. The continuous calculation regresses  $\log Q_h$  on centred regional wet-bulb temperature separately in each month-year record, then averages the slopes equally over months, scales and the 33 evaluation summers. The five-scale mean slope is  $-0.0650\text{ }^{\circ}\text{C}^{-1}$ , with a 95% interval of  $[-0.0822, -0.0479]$  and negative summer slopes in 31 of 33 years.

The calendar-matched calculation compares each high day with the mean of middle days no more than three or five calendar days away in the same month-year record. Eligible high-day contrasts are averaged within records before application of the original equal-month, equal-scale and equal-summer hierarchy. The  $\pm 3$ -day window matches 670 of 792 high days (84.6%); the  $\pm 5$ -day window matches 750 (94.7%). Each of the 99 evaluation month-year records contributes at least one match under both windows.

**Table S14.** Continuous-intensity and calendar-matched results for the 33 evaluation summers. Continuous entries are slopes of  $\log Q_h$  per  $^{\circ}\text{C}$ . Matching entries are percentage effects. Brackets give 95% intervals.

| Bandwidth       | Continuous slope [interval]    | $\pm 3$ -day match [interval] | $\pm 5$ -day match [interval] |
|-----------------|--------------------------------|-------------------------------|-------------------------------|
| 126             | $-0.0318$ $[-0.0435, -0.0201]$ | $-2.32$ $[-4.73, 0.09]$       | $-2.54$ $[-4.90, -0.18]$      |
| 252             | $-0.0408$ $[-0.0553, -0.0263]$ | $-3.29$ $[-6.03, -0.56]$      | $-3.54$ $[-6.18, -0.90]$      |
| 503             | $-0.0546$ $[-0.0735, -0.0356]$ | $-4.41$ $[-7.44, -1.38]$      | $-4.74$ $[-7.66, -1.83]$      |
| 1,006           | $-0.0877$ $[-0.1092, -0.0661]$ | $-7.17$ $[-10.32, -4.02]$     | $-7.90$ $[-10.94, -4.86]$     |
| 2,013           | $-0.1104$ $[-0.1325, -0.0882]$ | $-8.87$ $[-12.06, -5.68]$     | $-9.89$ $[-12.98, -6.80]$     |
| Five-scale mean | $-0.0650$ $[-0.0822, -0.0479]$ | $-5.21$ $[-8.01, -2.41]$      | $-5.72$ $[-8.43, -3.02]$      |

The matched effects are weaker than the unrestricted high-to-middle contrast, but their broad-to-local ordering remains. Calendar matching also changes the target population of day pairs: only high days with eligible middle days enter the high-day average.

## S10. Heavy-tailed ratio stress test

We compare the raw effect  $r - 1$ , the log effect  $\log r$  and the bounded symmetric effect  $2(r - 1)/(r + 1)$ , where  $r$  is the high-to-middle dispersion ratio. The raw ratio is the protocol-defined primary measure and is retained as such. It is asymmetric, can be unstable when the middle-day denominator is small, and gives different summaries when ratios are averaged rather than marginal means divided. Its randomisation distribution need not be centred at zero. The 495 observed middle-day denominators range from 2.25 to  $25.99\text{ }^{\circ}\text{C}^2$ ; the log ratio is the principal robustness summary. In the observed 33-summer record, the five-scale raw-ratio summary is  $-7.28\%$ , the back-transformed log-ratio summary is  $-8.30\%$ , and the bounded symmetric summary is  $-8.59\%$ .

The stress design uses the 121 observed site locations, exponential spatial covariance with 450-km range and 5% nugget, circular field correlation 0.45 and regional-mean autoregression 0.65. A common daily scale  $\sqrt{3/\chi_3^2}$  produces

heavy-tailed graph energy. Each scenario has 2,000 paired replications of six 30–31-day records. All three measures use the same 999 product shifts within a replication; the seed is 20260811.

**Table S15.** Raw, log and bounded effect measures under the heavy-tailed stress design. Bias and RMSE are on each measure's scale. Size and power are lower-tail rejection proportions at 0.05. Alternative targets correspond to a high-day energy multiplier of 0.93.

| Measure           | Null bias | Null RMSE | Size   | Alternative target | Alt. bias | Alt. RMSE | Power  |
|-------------------|-----------|-----------|--------|--------------------|-----------|-----------|--------|
| Raw ratio         | 0.3716    | 1.4725    | 0.0470 | −0.0700            | 0.3456    | 1.3694    | 0.0795 |
| Log ratio         | −0.0698   | 0.3283    | 0.0490 | −0.0726            | −0.0698   | 0.3283    | 0.0850 |
| Bounded symmetric | −0.0657   | 0.2771    | 0.0475 | −0.0725            | −0.0574   | 0.2748    | 0.0790 |

The three product-shift tests retain near-nominal size in this design. The raw ratio nevertheless has an RMSE more than five times that of the bounded measure under the null. Product-shift calibration concerns the randomisation distribution of a fixed statistic; ratio-estimation stability also depends on the right tail of daily energy. The log and bounded summaries reduce that instability, with the bounded measure giving the smallest RMSE in these cells. Good randomisation calibration therefore does not establish that the raw-ratio point estimator is stable. A log or symmetric scale would be a preferable default in future studies not constrained by this protocol.

## S11. Simulation specifications and results

### S11.1. Baseline day-level profile experiment

The baseline experiment uses the 121 observed coordinates and the five design graphs. Six records contain 30 or 31 days unless a null cell changes the record count or length. Regional means follow an AR(1) process with coefficient 0.65; type-7 within-record quartiles define high and middle days. Centred spatial anomalies have exponential covariance with 450-km range and 5% nugget, and temporal coefficient 0.45 unless changed below. Projection through  $\mathbf{I} - \mathbf{1}\mathbf{1}^T/121$  makes the anomaly mean exactly zero, so the field mean equals the series used for labels. The circular Gaussian nulls use wrap-around temporal covariance and satisfy record-wise cyclic invariance. The two finite-AR cells deliberately omit wrap-around and assess a boundary departure.

The alternatives condition their parameter changes on the high-day indicator. Amplitude multipliers are 0.97, 0.955, 0.94, 0.92 and 0.90. The archived range grid is labelled 525, 587, 650, 725 and 800 km, but the simulator uses three covariance matrices: both settings below 600 km use a 525-km range, both settings from 600 to below 750 km use a 650-km range, and the last uses 800 km. The near-duplicate weak and medium rows are retained in Table S17 because they form part of 25 archived cells, including two duplicated range implementations, not because they are distinct covariance designs. The range change enters the innovation covariance and can persist through the field AR(1) state. Gradient alternatives add the standardised pattern proportional to  $0.8x + y$  with middle-day amplitude 1.8 and high-day amplitudes 1.6, 1.5, 1.4, 1.3 and 1.2. Conditional targets are computed from the known quadratic forms rather than from Monte Carlo averages.

Seven tests use the same simulated fields, labels and sampled shifts. The graph test averages the five Gaussian-graph ratios; the variogram test averages five pair-distance-quintile ratios. The remaining tests use spatial variance, four-nearest-neighbour semivariance, 400–600-km binned semivariance, Moran's  $I$  and Geary's  $C$ . The first five are lower-tail tests; Moran and Geary are two-sided. Each of the 25 archived cells, including the two duplicated range implementations, contains 1,000 replications and 499 shifts. With 1,000 replications, the Monte Carlo standard error is 0.69 percentage points at 5% rejection and 1.58 points at 50% power.

These mechanisms were chosen by the authors and do not define a general optimality class. The graph profile has the largest minimum power across these 15 alternatives, but nearest-neighbour semivariance is stronger for range expansion and variance or the variogram profile is stronger for weak gradient suppression. Standardisation leaves Moran's  $I$  and Geary's  $C$  with near-null power for pure amplitude changes.



**Table S16.** Complete null calibration for the baseline day-level experiment. Entries are rejection percentages at 0.05 over 1,000 replications.

| Null cell                | Graph | Variogram | Variance | Nearest | 400–600 km | Moran | Geary |
|--------------------------|-------|-----------|----------|---------|------------|-------|-------|
| Circular baseline        | 4.6   | 4.9       | 5.4      | 5.5     | 4.3        | 4.7   | 5.2   |
| Circular, 3 records      | 6.8   | 6.1       | 6.0      | 5.8     | 5.2        | 5.7   | 5.0   |
| Circular, 12 records     | 5.3   | 5.7       | 5.9      | 4.4     | 4.8        | 4.9   | 4.2   |
| Circular, 20 days        | 5.8   | 5.3       | 5.0      | 5.2     | 6.4        | 4.8   | 4.8   |
| Circular, 60 days        | 6.2   | 5.7       | 5.9      | 5.9     | 6.0        | 6.1   | 6.1   |
| Circular, $\rho = 0$     | 6.2   | 5.8       | 5.6      | 4.0     | 5.7        | 5.6   | 5.2   |
| Circular, $\rho = 0.75$  | 4.5   | 4.2       | 4.1      | 5.4     | 4.1        | 4.9   | 4.1   |
| Circular, $t_3$ scale    | 6.1   | 5.6       | 5.6      | 6.1     | 5.9        | 5.0   | 6.1   |
| Finite AR baseline       | 4.6   | 4.2       | 4.4      | 4.6     | 4.6        | 5.5   | 4.9   |
| Finite AR, $\rho = 0.75$ | 5.1   | 5.1       | 5.0      | 4.6     | 5.1        | 4.7   | 5.8   |

**Table S17.** Complete power results for the baseline day-level alternatives. Target is the true five-graph contraction in per cent; other entries are rejection percentages at 0.05 over 1,000 replications.

| Mechanism | Setting                                       | Target | Graph | Variogram | Variance | Nearest | 400–600 km | Moran | Geary |
|-----------|-----------------------------------------------|--------|-------|-----------|----------|---------|------------|-------|-------|
| Amplitude | 0.970                                         | –5.91  | 50.0  | 33.6      | 31.1     | 76.4    | 45.0       | 5.2   | 5.9   |
| Amplitude | 0.955                                         | –8.80  | 80.5  | 57.3      | 54.4     | 96.9    | 75.1       | 5.0   | 5.0   |
| Amplitude | 0.940                                         | –11.64 | 95.3  | 79.8      | 77.1     | 99.8    | 91.9       | 4.6   | 4.5   |
| Amplitude | 0.920                                         | –15.36 | 99.7  | 93.3      | 91.1     | 100.0   | 99.0       | 4.4   | 3.8   |
| Amplitude | 0.900                                         | –19.00 | 99.9  | 99.3      | 99.1     | 100.0   | 99.9       | 5.4   | 4.7   |
| Range     | 525 km                                        | –6.07  | 48.9  | 19.0      | 16.4     | 95.4    | 51.5       | 15.5  | 15.9  |
| Range     | 587-km archived label (implemented as 525 km) | –6.07  | 50.8  | 20.3      | 17.8     | 95.2    | 52.4       | 13.2  | 15.1  |
| Range     | 650 km                                        | –14.38 | 98.6  | 62.4      | 53.3     | 100.0   | 99.1       | 45.3  | 53.7  |
| Range     | 725-km archived label (implemented as 650 km) | –14.39 | 98.8  | 60.8      | 51.0     | 100.0   | 99.6       | 49.0  | 56.2  |
| Range     | 800 km                                        | –22.20 | 100.0 | 91.4      | 85.6     | 100.0   | 100.0      | 74.0  | 85.0  |
| Gradient  | 1.6                                           | –10.34 | 90.9  | 93.9      | 96.5     | 40.8    | 78.9       | 32.4  | 81.6  |
| Gradient  | 1.5                                           | –15.06 | 99.6  | 99.9      | 99.9     | 65.2    | 95.0       | 64.8  | 98.7  |
| Gradient  | 1.4                                           | –19.47 | 100.0 | 100.0     | 100.0    | 82.5    | 99.8       | 87.6  | 100.0 |
| Gradient  | 1.3                                           | –23.57 | 100.0 | 100.0     | 100.0    | 94.5    | 100.0      | 98.0  | 100.0 |
| Gradient  | 1.2                                           | –27.37 | 100.0 | 100.0     | 100.0    | 97.6    | 100.0      | 99.8  | 100.0 |

## S11.2. Coupled label–field stress designs

The coupled simulation uses the 121 observed site coordinates and six records of lengths 30, 31, 30, 30, 31 and 30 days. Isotropic anomaly fields have exponential covariance with 450-km range and 5% nugget. The field state has autoregressive coefficient 0.45, and the regional weather series has coefficient 0.65. High and middle labels use type-7 quartiles within each record. A standardised spatial gradient is proportional to  $0.8x + y$ , where  $x$  and  $y$  are projected site coordinates. Its sign is driven by  $0.75\tilde{\mu}_t + (1 - 0.75^2)^{1/2}Z_t$ , so labels and spatial structure share a latent weather component even under the null.

**Table S18.** Exact changes defining the eight coupled label–field designs. Every cell retains the common six-record construction and within-record labels.

| Design                   | Null                                                                                                                                                           | Contraction alternative                                                   |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Shared latent gradient   | Gradient amplitude is 1.8 on every day; its sign shares the latent driver with regional weather.                                                               | Gradient amplitude on high days is reduced from 1.8 to 1.2.               |
| Within-month progression | Regional mean is a linear trend from –1.4 to 1.4 plus 0.60 times the AR(1) weather series; gradient amplitude remains 1.8.                                     | Gradient amplitude declines linearly from 1.95 to 1.30 across the record. |
| Anisotropic covariance   | Anomaly e-folding ranges are 800 km in projected longitude and 300 km in projected latitude.                                                                   | High-day anomaly amplitude is multiplied by 0.94.                         |
| Peak-hour selection      | Each daily regional mean is the maximum of 24 hourly values with sinusoidal amplitude 1.8 and independent noise SD 0.35; anomaly covariance is hour-invariant. | High-day anomaly amplitude is multiplied by 0.94.                         |

Each field is evaluated with the five Gaussian graphs and a five-bin variogram profile whose bins are the quintiles of the 7,260 observed pair distances. For each replication, the target is the high-to-middle contrast in the conditional expected quadratic forms. The graph and variogram statistics use the same simulated fields, labels and product shifts. Each of the eight cells contains 1,000 replications with 499 shifts per test.

### S11.3. Repeated-summer experiment

The annual experiment crosses  $R \in \{20, 33, 60, 120\}$ ,  $\rho \in \{0, 0.3, 0.6\}$ , Gaussian, centred log-normal and standardised  $t_3$  innovations, and effects  $\delta \in \{0, -0.035, -0.07\}$ . The resulting 108 cells each contain 10,000 replications after a 200-step burn-in. For retained summer  $r$ ,

$$U_r = \rho U_{r-1} + (1 - \rho^2)^{1/2} \varepsilon_r, \quad Y_r = \delta + 0.12 U_r.$$

The Gaussian innovation is standard normal. For  $Z \sim N(0, 1)$ , the centred unit-variance log-normal innovation is

$$\frac{\exp(0.65Z) - \exp(0.65^2/2)}{\{\{\exp(0.65^2) - 1\} \exp(0.65^2)\}^{1/2}},$$

and the heavy-tailed innovation is  $T_3/\sqrt{3}$ . The point estimate is  $\bar{Y}$ . The ordinary interval uses the sample variance and a  $t_{R-1}$  critical value. The lag-2 calculation uses a Bartlett long-run variance with the same critical value. The growing-lag interval uses  $L = \max\{1, \lfloor R^{1/3} \rfloor\}$  and a standard-normal critical value. The sign test uses the exact binomial upper tail of the number of negative summers. The sign calculation targets  $\Pr(Y_r < 0) = 1/2$ . The mean-null target is  $E(Y_r) = 0$ . Under the centred skewed innovations used below, the two targets can differ; sign-column rates in skewed mean-null cells therefore quantify rejection of the sign null. The three-test decision requires the Student, lag-2 and sign  $p$ -values all to be at most 0.025. Standardised diagnostics use the known long-run variance  $0.12^2(1 + \rho)/(1 - \rho)$ .

The following tables report all eight coupled cells and all 108 annual cells. Machine-readable versions of all 108 annual cells are provided in `results/year_inference_simulation_summary.csv` in the accompanying repository.

**Table S19.** Complete results for the eight coupled label–field stress designs. Targets, bias and RMSE are percentage points; rejection is per cent over 1,000 replications. Graph and five-bin variogram profiles use identical fields, labels and product shifts.

| Scenario                          | Family      | Graph profile |      |      |        | Five-bin variogram |      |      |        |
|-----------------------------------|-------------|---------------|------|------|--------|--------------------|------|------|--------|
|                                   |             | Target        | Bias | RMSE | Reject | Target             | Bias | RMSE | Reject |
| Shared-latent null                | Null        | 0.00          | 0.38 | 3.63 | 5.5    | -0.00              | 0.53 | 4.58 | 5.3    |
| Shared-latent contraction         | Alternative | -27.37        | 0.19 | 2.70 | 100.0  | -36.66             | 0.25 | 3.26 | 100.0  |
| Seasonal-progression null         | Null        | 0.00          | 0.64 | 3.76 | 4.0    | -0.00              | 0.85 | 4.77 | 4.3    |
| Seasonal-gradient contraction     | Alternative | -10.74        | 0.27 | 3.65 | 78.6   | -14.72             | 0.33 | 4.60 | 83.3   |
| Anisotropic null                  | Null        | 0.00          | 0.31 | 3.86 | 7.0    | 0.00               | 0.42 | 4.81 | 7.3    |
| Anisotropic amplitude contraction | Alternative | -5.92         | 0.21 | 3.63 | 51.4   | -3.98              | 0.32 | 4.58 | 25.5   |
| Peak-selection null               | Null        | 0.00          | 0.21 | 3.69 | 6.6    | -0.00              | 0.34 | 4.63 | 6.6    |
| Peak-selection contraction        | Alternative | -5.90         | 0.36 | 3.50 | 54.2   | -3.96              | 0.52 | 4.46 | 25.1   |

Table S20: Complete 108-cell repeated-summer results: estimation and coverage. Bias and RMSE are percentage points; coverage is per cent.

| $R$ | $\rho$ | Innovation | Effect | Bias<br>percentage points | RMSE<br>points | Student | Lag 2<br>coverage (%) | Growing |
|-----|--------|------------|--------|---------------------------|----------------|---------|-----------------------|---------|
| 20  | 0.0    | Gaussian   | Null   | 0.015                     | 2.655          | 95.4    | 91.7                  | 89.9    |
| 20  | 0.0    | Gaussian   | -3.5%  | -0.015                    | 2.680          | 94.8    | 91.1                  | 89.2    |
| 20  | 0.0    | Gaussian   | -7%    | -0.015                    | 2.665          | 95.1    | 91.3                  | 89.4    |
| 20  | 0.0    | Skewed     | Null   | -0.046                    | 2.691          | 91.2    | 87.6                  | 85.9    |
| 20  | 0.0    | Skewed     | -3.5%  | 0.022                     | 2.676          | 92.1    | 88.7                  | 86.9    |
| 20  | 0.0    | Skewed     | -7%    | -0.024                    | 2.662          | 91.8    | 88.3                  | 86.5    |
| 20  | 0.0    | $t_3$      | Null   | -0.028                    | 2.672          | 95.3    | 91.7                  | 89.7    |
| 20  | 0.0    | $t_3$      | -3.5%  | 0.028                     | 2.599          | 95.9    | 92.0                  | 90.4    |
| 20  | 0.0    | $t_3$      | -7%    | 0.028                     | 2.696          | 95.9    | 92.0                  | 90.1    |
| 20  | 0.3    | Gaussian   | Null   | 0.081                     | 3.584          | 85.5    | 86.8                  | 84.8    |
| 20  | 0.3    | Gaussian   | -3.5%  | -0.096                    | 3.597          | 85.0    | 86.6                  | 84.2    |
| 20  | 0.3    | Gaussian   | -7%    | 0.036                     | 3.620          | 85.0    | 86.5                  | 84.0    |
| 20  | 0.3    | Skewed     | Null   | 0.053                     | 3.612          | 82.7    | 84.4                  | 82.3    |
| 20  | 0.3    | Skewed     | -3.5%  | 0.019                     | 3.615          | 82.6    | 84.2                  | 82.2    |
| 20  | 0.3    | Skewed     | -7%    | -0.038                    | 3.545          | 83.0    | 84.6                  | 82.3    |
| 20  | 0.3    | $t_3$      | Null   | 0.056                     | 3.486          | 85.9    | 87.5                  | 85.1    |
| 20  | 0.3    | $t_3$      | -3.5%  | -0.126                    | 3.533          | 85.7    | 87.3                  | 84.8    |
| 20  | 0.3    | $t_3$      | -7%    | -0.002                    | 3.620          | 85.8    | 87.8                  | 85.3    |
| 20  | 0.6    | Gaussian   | Null   | -0.042                    | 5.096          | 66.5    | 76.2                  | 73.4    |
| 20  | 0.6    | Gaussian   | -3.5%  | 0.002                     | 5.128          | 67.0    | 76.3                  | 73.8    |
| 20  | 0.6    | Gaussian   | -7%    | 0.000                     | 5.175          | 66.6    | 75.9                  | 73.2    |
| 20  | 0.6    | Skewed     | Null   | 0.038                     | 5.187          | 64.4    | 73.9                  | 71.2    |
| 20  | 0.6    | Skewed     | -3.5%  | -0.043                    | 5.140          | 64.5    | 74.3                  | 71.5    |
| 20  | 0.6    | Skewed     | -7%    | -0.039                    | 5.115          | 65.1    | 74.6                  | 71.7    |
| 20  | 0.6    | $t_3$      | Null   | 0.047                     | 4.987          | 65.9    | 76.3                  | 73.5    |
| 20  | 0.6    | $t_3$      | -3.5%  | 0.075                     | 5.101          | 65.1    | 75.8                  | 72.7    |
| 20  | 0.6    | $t_3$      | -7%    | -0.026                    | 5.254          | 65.5    | 76.7                  | 73.7    |
| 33  | 0.0    | Gaussian   | Null   | 0.057                     | 2.065          | 95.2    | 93.3                  | 91.2    |
| 33  | 0.0    | Gaussian   | -3.5%  | -0.004                    | 2.078          | 95.2    | 93.1                  | 91.2    |
| 33  | 0.0    | Gaussian   | -7%    | -0.021                    | 2.062          | 95.2    | 93.5                  | 91.4    |
| 33  | 0.0    | Skewed     | Null   | -0.029                    | 2.071          | 92.4    | 90.6                  | 88.5    |
| 33  | 0.0    | Skewed     | -3.5%  | -0.002                    | 2.061          | 92.9    | 91.0                  | 89.0    |
| 33  | 0.0    | Skewed     | -7%    | 0.005                     | 2.080          | 92.6    | 90.7                  | 88.8    |
| 33  | 0.0    | $t_3$      | Null   | -0.019                    | 2.089          | 95.8    | 93.7                  | 91.5    |
| 33  | 0.0    | $t_3$      | -3.5%  | 0.007                     | 2.062          | 95.6    | 93.2                  | 91.0    |
| 33  | 0.0    | $t_3$      | -7%    | -0.004                    | 2.127          | 95.7    | 93.6                  | 91.5    |
| 33  | 0.3    | Gaussian   | Null   | -0.046                    | 2.837          | 85.4    | 89.0                  | 87.3    |
| 33  | 0.3    | Gaussian   | -3.5%  | -0.002                    | 2.819          | 85.4    | 88.6                  | 87.2    |
| 33  | 0.3    | Gaussian   | -7%    | 0.018                     | 2.808          | 85.7    | 88.8                  | 87.3    |
| 33  | 0.3    | Skewed     | Null   | 0.000                     | 2.795          | 83.3    | 87.2                  | 85.9    |
| 33  | 0.3    | Skewed     | -3.5%  | -0.004                    | 2.834          | 82.9    | 86.4                  | 85.1    |
| 33  | 0.3    | Skewed     | -7%    | -0.008                    | 2.792          | 83.4    | 86.8                  | 85.4    |
| 33  | 0.3    | $t_3$      | Null   | -0.011                    | 2.773          | 85.0    | 89.2                  | 87.5    |
| 33  | 0.3    | $t_3$      | -3.5%  | 0.012                     | 2.804          | 85.5    | 89.4                  | 88.2    |
| 33  | 0.3    | $t_3$      | -7%    | 0.033                     | 2.760          | 85.4    | 89.1                  | 87.8    |
| 33  | 0.6    | Gaussian   | Null   | 0.004                     | 4.112          | 66.1    | 78.7                  | 79.0    |
| 33  | 0.6    | Gaussian   | -3.5%  | 0.031                     | 4.059          | 66.8    | 79.1                  | 79.2    |
| 33  | 0.6    | Gaussian   | -7%    | 0.027                     | 4.049          | 67.3    | 79.5                  | 79.6    |
| 33  | 0.6    | Skewed     | Null   | 0.020                     | 4.090          | 64.9    | 77.6                  | 77.9    |
| 33  | 0.6    | Skewed     | -3.5%  | 0.043                     | 4.053          | 65.6    | 77.8                  | 78.0    |
| 33  | 0.6    | Skewed     | -7%    | 0.037                     | 4.070          | 65.0    | 77.7                  | 77.7    |
| 33  | 0.6    | $t_3$      | Null   | -0.014                    | 3.994          | 66.5    | 79.8                  | 80.1    |
| 33  | 0.6    | $t_3$      | -3.5%  | -0.054                    | 4.004          | 66.0    | 79.9                  | 80.3    |
| 33  | 0.6    | $t_3$      | -7%    | 0.006                     | 4.158          | 65.4    | 79.6                  | 79.8    |
| 60  | 0.0    | Gaussian   | Null   | 0.002                     | 1.554          | 95.1    | 94.0                  | 92.9    |
| 60  | 0.0    | Gaussian   | -3.5%  | -0.018                    | 1.562          | 94.8    | 93.6                  | 92.4    |

Continued on next page

Table S20 continued

| $R$ | $\rho$ | Innovation | Effect | Bias   | RMSE  | Student | Lag 2 | Growing |
|-----|--------|------------|--------|--------|-------|---------|-------|---------|
| 60  | 0.0    | Gaussian   | -7%    | -0.003 | 1.551 | 95.0    | 93.8  | 92.8    |
| 60  | 0.0    | Skewed     | Null   | -0.002 | 1.553 | 93.4    | 92.2  | 91.1    |
| 60  | 0.0    | Skewed     | -3.5%  | 0.020  | 1.541 | 93.5    | 92.5  | 91.4    |
| 60  | 0.0    | Skewed     | -7%    | 0.039  | 1.545 | 94.1    | 93.2  | 92.0    |
| 60  | 0.0    | $t_3$      | Null   | 0.019  | 1.583 | 95.4    | 94.3  | 93.2    |
| 60  | 0.0    | $t_3$      | -3.5%  | 0.008  | 1.523 | 95.5    | 94.3  | 93.3    |
| 60  | 0.0    | $t_3$      | -7%    | -0.008 | 1.536 | 95.7    | 94.8  | 93.6    |
| 60  | 0.3    | Gaussian   | Null   | 0.016  | 2.097 | 85.0    | 90.0  | 89.7    |
| 60  | 0.3    | Gaussian   | -3.5%  | -0.016 | 2.068 | 85.9    | 90.9  | 90.4    |
| 60  | 0.3    | Gaussian   | -7%    | 0.033  | 2.083 | 85.6    | 90.6  | 90.3    |
| 60  | 0.3    | Skewed     | Null   | -0.019 | 2.085 | 84.2    | 89.0  | 88.8    |
| 60  | 0.3    | Skewed     | -3.5%  | 0.018  | 2.096 | 84.1    | 89.1  | 88.8    |
| 60  | 0.3    | Skewed     | -7%    | -0.001 | 2.097 | 83.7    | 88.6  | 88.4    |
| 60  | 0.3    | $t_3$      | Null   | -0.014 | 2.082 | 85.2    | 90.5  | 90.0    |
| 60  | 0.3    | $t_3$      | -3.5%  | 0.035  | 2.088 | 84.7    | 90.7  | 90.3    |
| 60  | 0.3    | $t_3$      | -7%    | -0.010 | 2.087 | 85.1    | 90.5  | 90.1    |
| 60  | 0.6    | Gaussian   | Null   | -0.011 | 3.056 | 67.0    | 81.0  | 82.5    |
| 60  | 0.6    | Gaussian   | -3.5%  | -0.029 | 3.066 | 66.9    | 81.5  | 83.0    |
| 60  | 0.6    | Gaussian   | -7%    | 0.030  | 3.036 | 67.3    | 81.3  | 82.8    |
| 60  | 0.6    | Skewed     | Null   | 0.009  | 3.026 | 65.9    | 80.1  | 81.5    |
| 60  | 0.6    | Skewed     | -3.5%  | -0.016 | 3.062 | 65.4    | 79.9  | 81.5    |
| 60  | 0.6    | Skewed     | -7%    | -0.018 | 3.038 | 66.5    | 80.2  | 81.6    |
| 60  | 0.6    | $t_3$      | Null   | -0.003 | 2.988 | 66.1    | 81.0  | 82.6    |
| 60  | 0.6    | $t_3$      | -3.5%  | 0.005  | 3.063 | 66.6    | 81.3  | 82.9    |
| 60  | 0.6    | $t_3$      | -7%    | 0.026  | 3.046 | 66.0    | 81.5  | 83.0    |
| 120 | 0.0    | Gaussian   | Null   | 0.013  | 1.098 | 94.8    | 94.3  | 93.7    |
| 120 | 0.0    | Gaussian   | -3.5%  | 0.003  | 1.100 | 94.9    | 94.4  | 93.7    |
| 120 | 0.0    | Gaussian   | -7%    | 0.023  | 1.101 | 94.7    | 94.2  | 93.2    |
| 120 | 0.0    | Skewed     | Null   | 0.011  | 1.091 | 94.4    | 93.7  | 93.0    |
| 120 | 0.0    | Skewed     | -3.5%  | 0.003  | 1.084 | 94.2    | 93.5  | 93.1    |
| 120 | 0.0    | Skewed     | -7%    | -0.004 | 1.101 | 93.7    | 93.3  | 92.5    |
| 120 | 0.0    | $t_3$      | Null   | 0.008  | 1.115 | 95.2    | 94.5  | 93.9    |
| 120 | 0.0    | $t_3$      | -3.5%  | -0.007 | 1.099 | 95.2    | 94.4  | 93.8    |
| 120 | 0.0    | $t_3$      | -7%    | 0.004  | 1.100 | 95.5    | 94.8  | 94.0    |
| 120 | 0.3    | Gaussian   | Null   | 0.013  | 1.493 | 84.8    | 90.8  | 91.5    |
| 120 | 0.3    | Gaussian   | -3.5%  | 0.001  | 1.471 | 85.6    | 91.2  | 91.8    |
| 120 | 0.3    | Gaussian   | -7%    | 0.002  | 1.489 | 84.8    | 90.9  | 91.5    |
| 120 | 0.3    | Skewed     | Null   | -0.004 | 1.502 | 83.6    | 89.6  | 90.2    |
| 120 | 0.3    | Skewed     | -3.5%  | -0.014 | 1.485 | 84.7    | 90.2  | 90.7    |
| 120 | 0.3    | Skewed     | -7%    | 0.025  | 1.491 | 84.2    | 90.0  | 90.4    |
| 120 | 0.3    | $t_3$      | Null   | -0.009 | 1.495 | 84.7    | 91.1  | 91.7    |
| 120 | 0.3    | $t_3$      | -3.5%  | -0.028 | 1.475 | 84.6    | 91.2  | 91.7    |
| 120 | 0.3    | $t_3$      | -7%    | -0.002 | 1.488 | 84.5    | 90.7  | 91.4    |
| 120 | 0.6    | Gaussian   | Null   | -0.018 | 2.164 | 67.9    | 82.8  | 86.5    |
| 120 | 0.6    | Gaussian   | -3.5%  | -0.010 | 2.171 | 67.8    | 82.7  | 86.6    |
| 120 | 0.6    | Gaussian   | -7%    | 0.023  | 2.169 | 66.9    | 82.3  | 86.5    |
| 120 | 0.6    | Skewed     | Null   | -0.012 | 2.162 | 67.2    | 81.8  | 85.6    |
| 120 | 0.6    | Skewed     | -3.5%  | -0.020 | 2.181 | 65.7    | 81.3  | 85.6    |
| 120 | 0.6    | Skewed     | -7%    | 0.014  | 2.180 | 66.5    | 81.6  | 85.5    |
| 120 | 0.6    | $t_3$      | Null   | -0.013 | 2.184 | 66.6    | 82.5  | 86.4    |
| 120 | 0.6    | $t_3$      | -3.5%  | -0.053 | 2.146 | 66.5    | 82.5  | 86.6    |
| 120 | 0.6    | $t_3$      | -7%    | 0.015  | 2.180 | 65.7    | 82.2  | 86.3    |

Table S21: Complete 108-cell repeated-summer results: standardised-mean diagnostics.  $L$  is the growing Bartlett lag; the standardised columns use the known long-run variance.

| $R$ | $\rho$ | Innovation | Effect | $L$ | $z$ mean | $z$ SD | $z_{.025}, z_{.975}$ |
|-----|--------|------------|--------|-----|----------|--------|----------------------|
| 20  | 0.0    | Gaussian   | Null   | 2   | 0.006    | 0.990  | -1.94, 1.98          |
| 20  | 0.0    | Gaussian   | -3.5%  | 2   | -0.006   | 0.999  | -1.98, 1.98          |
| 20  | 0.0    | Gaussian   | -7%    | 2   | -0.006   | 0.993  | -1.92, 1.99          |
| 20  | 0.0    | Skewed     | Null   | 2   | -0.017   | 1.003  | -1.71, 2.21          |
| 20  | 0.0    | Skewed     | -3.5%  | 2   | 0.008    | 0.997  | -1.71, 2.18          |
| 20  | 0.0    | Skewed     | -7%    | 2   | -0.009   | 0.992  | -1.68, 2.14          |
| 20  | 0.0    | $t_3$      | Null   | 2   | -0.010   | 0.996  | -1.92, 1.95          |
| 20  | 0.0    | $t_3$      | -3.5%  | 2   | 0.011    | 0.968  | -1.90, 1.90          |
| 20  | 0.0    | $t_3$      | -7%    | 2   | 0.010    | 1.005  | -1.88, 1.91          |
| 20  | 0.3    | Gaussian   | Null   | 2   | 0.022    | 0.980  | -1.91, 1.94          |
| 20  | 0.3    | Gaussian   | -3.5%  | 2   | -0.026   | 0.983  | -1.96, 1.93          |
| 20  | 0.3    | Gaussian   | -7%    | 2   | 0.010    | 0.990  | -1.93, 1.92          |
| 20  | 0.3    | Skewed     | Null   | 2   | 0.014    | 0.988  | -1.66, 2.18          |
| 20  | 0.3    | Skewed     | -3.5%  | 2   | 0.005    | 0.989  | -1.68, 2.17          |
| 20  | 0.3    | Skewed     | -7%    | 2   | -0.010   | 0.969  | -1.68, 2.12          |
| 20  | 0.3    | $t_3$      | Null   | 2   | 0.015    | 0.953  | -1.85, 1.91          |
| 20  | 0.3    | $t_3$      | -3.5%  | 2   | -0.035   | 0.966  | -1.94, 1.85          |
| 20  | 0.3    | $t_3$      | -7%    | 2   | -0.001   | 0.990  | -1.92, 1.91          |
| 20  | 0.6    | Gaussian   | Null   | 2   | -0.008   | 0.950  | -1.89, 1.83          |
| 20  | 0.6    | Gaussian   | -3.5%  | 2   | 0.000    | 0.956  | -1.89, 1.85          |
| 20  | 0.6    | Gaussian   | -7%    | 2   | 0.000    | 0.964  | -1.91, 1.88          |
| 20  | 0.6    | Skewed     | Null   | 2   | 0.007    | 0.967  | -1.64, 2.10          |
| 20  | 0.6    | Skewed     | -3.5%  | 2   | -0.008   | 0.958  | -1.62, 2.11          |
| 20  | 0.6    | Skewed     | -7%    | 2   | -0.007   | 0.953  | -1.60, 2.11          |
| 20  | 0.6    | $t_3$      | Null   | 2   | 0.009    | 0.929  | -1.77, 1.81          |
| 20  | 0.6    | $t_3$      | -3.5%  | 2   | 0.014    | 0.950  | -1.81, 1.87          |
| 20  | 0.6    | $t_3$      | -7%    | 2   | -0.005   | 0.979  | -1.84, 1.79          |
| 33  | 0.0    | Gaussian   | Null   | 3   | 0.027    | 0.988  | -1.92, 1.97          |
| 33  | 0.0    | Gaussian   | -3.5%  | 3   | -0.002   | 0.995  | -1.95, 1.92          |
| 33  | 0.0    | Gaussian   | -7%    | 3   | -0.010   | 0.987  | -1.93, 1.93          |
| 33  | 0.0    | Skewed     | Null   | 3   | -0.014   | 0.991  | -1.75, 2.12          |
| 33  | 0.0    | Skewed     | -3.5%  | 3   | -0.001   | 0.987  | -1.72, 2.11          |
| 33  | 0.0    | Skewed     | -7%    | 3   | 0.002    | 0.996  | -1.73, 2.13          |
| 33  | 0.0    | $t_3$      | Null   | 3   | -0.009   | 1.000  | -1.93, 1.92          |
| 33  | 0.0    | $t_3$      | -3.5%  | 3   | 0.003    | 0.987  | -1.93, 1.93          |
| 33  | 0.0    | $t_3$      | -7%    | 3   | -0.002   | 1.018  | -1.94, 1.91          |
| 33  | 0.3    | Gaussian   | Null   | 3   | -0.016   | 0.996  | -1.99, 1.95          |
| 33  | 0.3    | Gaussian   | -3.5%  | 3   | -0.001   | 0.990  | -1.93, 1.94          |
| 33  | 0.3    | Gaussian   | -7%    | 3   | 0.006    | 0.987  | -1.92, 1.92          |
| 33  | 0.3    | Skewed     | Null   | 3   | 0.000    | 0.982  | -1.72, 2.11          |
| 33  | 0.3    | Skewed     | -3.5%  | 3   | -0.001   | 0.995  | -1.71, 2.13          |
| 33  | 0.3    | Skewed     | -7%    | 3   | -0.003   | 0.981  | -1.73, 2.09          |
| 33  | 0.3    | $t_3$      | Null   | 3   | -0.004   | 0.974  | -1.93, 1.89          |
| 33  | 0.3    | $t_3$      | -3.5%  | 3   | 0.004    | 0.985  | -1.92, 1.94          |
| 33  | 0.3    | $t_3$      | -7%    | 3   | 0.012    | 0.970  | -1.94, 1.91          |
| 33  | 0.6    | Gaussian   | Null   | 3   | 0.001    | 0.984  | -1.92, 1.92          |
| 33  | 0.6    | Gaussian   | -3.5%  | 3   | 0.008    | 0.972  | -1.89, 1.87          |
| 33  | 0.6    | Gaussian   | -7%    | 3   | 0.006    | 0.969  | -1.90, 1.88          |
| 33  | 0.6    | Skewed     | Null   | 3   | 0.005    | 0.979  | -1.70, 2.12          |
| 33  | 0.6    | Skewed     | -3.5%  | 3   | 0.010    | 0.970  | -1.68, 2.10          |
| 33  | 0.6    | Skewed     | -7%    | 3   | 0.009    | 0.974  | -1.68, 2.09          |
| 33  | 0.6    | $t_3$      | Null   | 3   | -0.003   | 0.956  | -1.87, 1.83          |
| 33  | 0.6    | $t_3$      | -3.5%  | 3   | -0.013   | 0.958  | -1.89, 1.82          |
| 33  | 0.6    | $t_3$      | -7%    | 3   | 0.001    | 0.995  | -1.96, 1.88          |
| 60  | 0.0    | Gaussian   | Null   | 3   | 0.001    | 1.003  | -1.93, 1.95          |
| 60  | 0.0    | Gaussian   | -3.5%  | 3   | -0.012   | 1.008  | -2.01, 1.94          |
| 60  | 0.0    | Gaussian   | -7%    | 3   | -0.002   | 1.001  | -1.94, 1.98          |

Continued on next page

Table S21 continued

| $R$ | $\rho$ | Innovation | Effect | $L$ | $z$ mean | $z$ SD | $z_{.025}, z_{.975}$ |
|-----|--------|------------|--------|-----|----------|--------|----------------------|
| 60  | 0.0    | Skewed     | Null   | 3   | -0.001   | 1.003  | -1.79, 2.10          |
| 60  | 0.0    | Skewed     | -3.5%  | 3   | 0.013    | 0.995  | -1.79, 2.09          |
| 60  | 0.0    | Skewed     | -7%    | 3   | 0.025    | 0.997  | -1.78, 2.15          |
| 60  | 0.0    | $t_3$      | Null   | 3   | 0.013    | 1.022  | -1.96, 2.01          |
| 60  | 0.0    | $t_3$      | -3.5%  | 3   | 0.005    | 0.983  | -1.92, 1.96          |
| 60  | 0.0    | $t_3$      | -7%    | 3   | -0.005   | 0.991  | -1.89, 1.92          |
| 60  | 0.3    | Gaussian   | Null   | 3   | 0.008    | 0.993  | -1.99, 1.92          |
| 60  | 0.3    | Gaussian   | -3.5%  | 3   | -0.007   | 0.979  | -1.92, 1.90          |
| 60  | 0.3    | Gaussian   | -7%    | 3   | 0.015    | 0.986  | -1.91, 1.91          |
| 60  | 0.3    | Skewed     | Null   | 3   | -0.009   | 0.988  | -1.81, 2.07          |
| 60  | 0.3    | Skewed     | -3.5%  | 3   | 0.008    | 0.993  | -1.78, 2.13          |
| 60  | 0.3    | Skewed     | -7%    | 3   | -0.001   | 0.993  | -1.83, 2.08          |
| 60  | 0.3    | $t_3$      | Null   | 3   | -0.007   | 0.986  | -1.92, 1.98          |
| 60  | 0.3    | $t_3$      | -3.5%  | 3   | 0.017    | 0.989  | -1.91, 1.96          |
| 60  | 0.3    | $t_3$      | -7%    | 3   | -0.005   | 0.988  | -1.96, 1.95          |
| 60  | 0.6    | Gaussian   | Null   | 3   | -0.003   | 0.986  | -1.92, 1.94          |
| 60  | 0.6    | Gaussian   | -3.5%  | 3   | -0.009   | 0.989  | -1.96, 1.90          |
| 60  | 0.6    | Gaussian   | -7%    | 3   | 0.010    | 0.980  | -1.89, 1.92          |
| 60  | 0.6    | Skewed     | Null   | 3   | 0.003    | 0.977  | -1.78, 2.06          |
| 60  | 0.6    | Skewed     | -3.5%  | 3   | -0.005   | 0.988  | -1.77, 2.10          |
| 60  | 0.6    | Skewed     | -7%    | 3   | -0.006   | 0.980  | -1.76, 2.05          |
| 60  | 0.6    | $t_3$      | Null   | 3   | -0.001   | 0.964  | -1.90, 1.87          |
| 60  | 0.6    | $t_3$      | -3.5%  | 3   | 0.002    | 0.989  | -1.89, 1.92          |
| 60  | 0.6    | $t_3$      | -7%    | 3   | 0.008    | 0.983  | -1.94, 1.92          |
| 120 | 0.0    | Gaussian   | Null   | 4   | 0.012    | 1.003  | -1.98, 1.97          |
| 120 | 0.0    | Gaussian   | -3.5%  | 4   | 0.002    | 1.004  | -1.93, 1.99          |
| 120 | 0.0    | Gaussian   | -7%    | 4   | 0.021    | 1.005  | -1.94, 2.02          |
| 120 | 0.0    | Skewed     | Null   | 4   | 0.010    | 0.996  | -1.82, 2.07          |
| 120 | 0.0    | Skewed     | -3.5%  | 4   | 0.003    | 0.989  | -1.86, 2.05          |
| 120 | 0.0    | Skewed     | -7%    | 4   | -0.003   | 1.005  | -1.88, 2.07          |
| 120 | 0.0    | $t_3$      | Null   | 4   | 0.007    | 1.018  | -1.95, 2.01          |
| 120 | 0.0    | $t_3$      | -3.5%  | 4   | -0.006   | 1.003  | -1.96, 1.96          |
| 120 | 0.0    | $t_3$      | -7%    | 4   | 0.004    | 1.004  | -1.92, 2.02          |
| 120 | 0.3    | Gaussian   | Null   | 4   | 0.008    | 1.000  | -1.98, 1.95          |
| 120 | 0.3    | Gaussian   | -3.5%  | 4   | 0.001    | 0.986  | -1.92, 1.94          |
| 120 | 0.3    | Gaussian   | -7%    | 4   | 0.001    | 0.997  | -1.94, 1.95          |
| 120 | 0.3    | Skewed     | Null   | 4   | -0.003   | 1.006  | -1.86, 2.10          |
| 120 | 0.3    | Skewed     | -3.5%  | 4   | -0.009   | 0.995  | -1.87, 2.06          |
| 120 | 0.3    | Skewed     | -7%    | 4   | 0.017    | 0.999  | -1.82, 2.06          |
| 120 | 0.3    | $t_3$      | Null   | 4   | -0.006   | 1.001  | -1.94, 1.93          |
| 120 | 0.3    | $t_3$      | -3.5%  | 4   | -0.018   | 0.988  | -1.95, 1.94          |
| 120 | 0.3    | $t_3$      | -7%    | 4   | -0.001   | 0.997  | -1.98, 1.94          |
| 120 | 0.6    | Gaussian   | Null   | 4   | -0.008   | 0.988  | -1.97, 1.95          |
| 120 | 0.6    | Gaussian   | -3.5%  | 4   | -0.005   | 0.991  | -1.97, 1.94          |
| 120 | 0.6    | Gaussian   | -7%    | 4   | 0.010    | 0.990  | -1.92, 1.93          |
| 120 | 0.6    | Skewed     | Null   | 4   | -0.005   | 0.987  | -1.83, 2.09          |
| 120 | 0.6    | Skewed     | -3.5%  | 4   | -0.009   | 0.996  | -1.85, 2.06          |
| 120 | 0.6    | Skewed     | -7%    | 4   | 0.006    | 0.995  | -1.82, 2.08          |
| 120 | 0.6    | $t_3$      | Null   | 4   | -0.006   | 0.997  | -1.87, 1.91          |
| 120 | 0.6    | $t_3$      | -3.5%  | 4   | -0.024   | 0.979  | -1.97, 1.90          |
| 120 | 0.6    | $t_3$      | -7%    | 4   | 0.007    | 0.995  | -1.95, 1.93          |

Table S22: Complete 108-cell repeated-summer results: one-sided rejection percentages. For Student, lag-2 and growing-lag tests, null rows estimate mean-null size and non-null rows estimate power. The sign calculation targets  $\Pr(Y_r < 0) = 1/2$ ; under centred skewed mean-null innovations, its rejection rate quantifies departure from the sign null. The three-test column requires Student, lag-2 and sign values all to be at most 0.025.

| $R$ | $\rho$ | Innovation | Effect | Student | Lag 2 | Growing | Sign | Three-test |
|-----|--------|------------|--------|---------|-------|---------|------|------------|
| 20  | 0.0    | Gaussian   | Null   | 4.6     | 7.1   | 8.0     | 2.1  | 1.0        |
| 20  | 0.0    | Gaussian   | −3.5%  | 34.8    | 40.7  | 43.6    | 15.8 | 12.1       |
| 20  | 0.0    | Gaussian   | −7%    | 81.0    | 84.3  | 86.2    | 49.5 | 46.4       |
| 20  | 0.0    | Skewed     | Null   | 11.8    | 14.6  | 15.6    | 19.0 | 6.3        |
| 20  | 0.0    | Skewed     | −3.5%  | 45.4    | 49.5  | 51.6    | 53.8 | 31.2       |
| 20  | 0.0    | Skewed     | −7%    | 77.3    | 80.3  | 81.5    | 82.9 | 64.7       |
| 20  | 0.0    | $t_3$      | Null   | 5.2     | 7.5   | 8.4     | 2.2  | 0.9        |
| 20  | 0.0    | $t_3$      | −3.5%  | 43.9    | 49.8  | 52.3    | 32.7 | 21.2       |
| 20  | 0.0    | $t_3$      | −7%    | 85.7    | 87.7  | 88.8    | 82.8 | 71.0       |
| 20  | 0.3    | Gaussian   | Null   | 10.5    | 9.3   | 10.4    | 4.8  | 3.1        |
| 20  | 0.3    | Gaussian   | −3.5%  | 41.0    | 37.6  | 39.9    | 21.9 | 17.9       |
| 20  | 0.3    | Gaussian   | −7%    | 74.5    | 70.5  | 72.6    | 51.0 | 46.9       |
| 20  | 0.3    | Skewed     | Null   | 17.4    | 16.5  | 17.6    | 20.3 | 10.3       |
| 20  | 0.3    | Skewed     | −3.5%  | 47.8    | 44.6  | 46.3    | 50.9 | 33.3       |
| 20  | 0.3    | Skewed     | −7%    | 75.2    | 72.7  | 74.1    | 77.2 | 62.1       |
| 20  | 0.3    | $t_3$      | Null   | 11.3    | 10.0  | 11.0    | 5.5  | 3.1        |
| 20  | 0.3    | $t_3$      | −3.5%  | 48.8    | 44.7  | 46.9    | 35.2 | 25.9       |
| 20  | 0.3    | $t_3$      | −7%    | 81.3    | 78.0  | 79.7    | 75.4 | 64.6       |
| 20  | 0.6    | Gaussian   | Null   | 20.8    | 15.9  | 17.1    | 11.8 | 9.3        |
| 20  | 0.6    | Gaussian   | −3.5%  | 44.4    | 36.0  | 37.7    | 28.7 | 24.2       |
| 20  | 0.6    | Gaussian   | −7%    | 70.2    | 62.0  | 63.8    | 52.5 | 47.9       |
| 20  | 0.6    | Skewed     | Null   | 27.0    | 22.1  | 23.0    | 24.1 | 16.9       |
| 20  | 0.6    | Skewed     | −3.5%  | 51.7    | 44.9  | 46.5    | 47.2 | 36.5       |
| 20  | 0.6    | Skewed     | −7%    | 71.8    | 65.3  | 66.7    | 68.5 | 57.1       |
| 20  | 0.6    | $t_3$      | Null   | 21.5    | 16.0  | 17.2    | 12.6 | 9.4        |
| 20  | 0.6    | $t_3$      | −3.5%  | 49.2    | 40.8  | 42.4    | 36.9 | 29.6       |
| 20  | 0.6    | $t_3$      | −7%    | 76.2    | 68.8  | 70.4    | 66.9 | 58.6       |
| 33  | 0.0    | Gaussian   | Null   | 4.8     | 6.0   | 7.1     | 3.6  | 0.8        |
| 33  | 0.0    | Gaussian   | −3.5%  | 49.6    | 53.2  | 56.3    | 33.8 | 18.1       |
| 33  | 0.0    | Gaussian   | −7%    | 95.0    | 95.6  | 96.1    | 81.4 | 68.5       |
| 33  | 0.0    | Skewed     | Null   | 10.3    | 12.0  | 13.3    | 39.3 | 5.9        |
| 33  | 0.0    | Skewed     | −3.5%  | 55.5    | 58.1  | 60.6    | 84.7 | 42.7       |
| 33  | 0.0    | Skewed     | −7%    | 89.0    | 89.8  | 90.8    | 98.4 | 81.8       |
| 33  | 0.0    | $t_3$      | Null   | 5.0     | 6.5   | 7.8     | 4.0  | 0.7        |
| 33  | 0.0    | $t_3$      | −3.5%  | 58.2    | 61.0  | 63.6    | 62.0 | 33.2       |
| 33  | 0.0    | $t_3$      | −7%    | 94.4    | 94.9  | 95.6    | 98.3 | 87.9       |
| 33  | 0.3    | Gaussian   | Null   | 11.8    | 9.1   | 9.9     | 8.2  | 3.0        |
| 33  | 0.3    | Gaussian   | −3.5%  | 50.5    | 44.1  | 45.3    | 37.0 | 23.0       |
| 33  | 0.3    | Gaussian   | −7%    | 89.2    | 85.1  | 85.5    | 77.5 | 63.8       |
| 33  | 0.3    | Skewed     | Null   | 17.1    | 14.4  | 14.9    | 36.5 | 9.4        |
| 33  | 0.3    | Skewed     | −3.5%  | 55.7    | 50.7  | 51.6    | 76.8 | 40.7       |
| 33  | 0.3    | Skewed     | −7%    | 85.2    | 81.8  | 82.3    | 95.2 | 74.7       |
| 33  | 0.3    | $t_3$      | Null   | 11.7    | 9.3   | 10.0    | 9.1  | 3.3        |
| 33  | 0.3    | $t_3$      | −3.5%  | 57.8    | 51.2  | 52.2    | 57.5 | 33.7       |
| 33  | 0.3    | $t_3$      | −7%    | 90.7    | 87.9  | 88.2    | 94.3 | 79.5       |
| 33  | 0.6    | Gaussian   | Null   | 21.2    | 14.4  | 14.1    | 15.9 | 8.5        |
| 33  | 0.6    | Gaussian   | −3.5%  | 50.8    | 40.2  | 39.4    | 41.1 | 27.8       |
| 33  | 0.6    | Gaussian   | −7%    | 80.7    | 72.4  | 71.4    | 71.5 | 59.2       |
| 33  | 0.6    | Skewed     | Null   | 26.1    | 20.0  | 19.5    | 34.3 | 14.9       |
| 33  | 0.6    | Skewed     | −3.5%  | 55.5    | 46.7  | 45.9    | 65.2 | 39.0       |
| 33  | 0.6    | Skewed     | −7%    | 79.5    | 72.0  | 71.2    | 85.6 | 65.1       |
| 33  | 0.6    | $t_3$      | Null   | 21.6    | 14.4  | 14.1    | 16.5 | 8.1        |
| 33  | 0.6    | $t_3$      | −3.5%  | 56.9    | 46.7  | 45.8    | 52.6 | 34.8       |
| 33  | 0.6    | $t_3$      | −7%    | 84.8    | 77.6  | 76.8    | 84.6 | 68.4       |
| 60  | 0.0    | Gaussian   | Null   | 5.0     | 5.8   | 6.5     | 4.9  | 0.8        |

Continued on next page

Table S22 continued

| $R$ | $\rho$ | Innovation | Effect | Student | Lag 2 | Growing | Sign  | Three-test |
|-----|--------|------------|--------|---------|-------|---------|-------|------------|
| 60  | 0.0    | Gaussian   | −3.5%  | 71.8    | 73.6  | 74.7    | 54.4  | 31.7       |
| 60  | 0.0    | Gaussian   | −7%    | 99.7    | 99.7  | 99.7    | 96.8  | 90.7       |
| 60  | 0.0    | Skewed     | Null   | 9.2     | 10.0  | 10.6    | 62.6  | 5.2        |
| 60  | 0.0    | Skewed     | −3.5%  | 70.6    | 71.7  | 72.7    | 98.3  | 59.5       |
| 60  | 0.0    | Skewed     | −7%    | 97.6    | 97.7  | 97.9    | 100.0 | 95.5       |
| 60  | 0.0    | $t_3$      | Null   | 4.7     | 5.5   | 6.2     | 4.7   | 0.5        |
| 60  | 0.0    | $t_3$      | −3.5%  | 77.5    | 78.2  | 79.1    | 87.0  | 57.0       |
| 60  | 0.0    | $t_3$      | −7%    | 98.8    | 98.7  | 98.8    | 100.0 | 97.6       |
| 60  | 0.3    | Gaussian   | Null   | 11.2    | 8.2   | 8.3     | 8.9   | 2.8        |
| 60  | 0.3    | Gaussian   | −3.5%  | 67.5    | 59.5  | 59.4    | 54.2  | 33.7       |
| 60  | 0.3    | Gaussian   | −7%    | 98.4    | 97.0  | 96.9    | 93.9  | 86.0       |
| 60  | 0.3    | Skewed     | Null   | 15.2    | 11.9  | 12.1    | 52.5  | 8.3        |
| 60  | 0.3    | Skewed     | −3.5%  | 67.5    | 61.3  | 61.1    | 93.7  | 52.2       |
| 60  | 0.3    | Skewed     | −7%    | 95.5    | 93.2  | 93.1    | 99.7  | 89.6       |
| 60  | 0.3    | $t_3$      | Null   | 11.8    | 8.1   | 8.1     | 10.0  | 2.4        |
| 60  | 0.3    | $t_3$      | −3.5%  | 71.9    | 64.7  | 64.5    | 77.5  | 47.9       |
| 60  | 0.3    | $t_3$      | −7%    | 97.4    | 96.5  | 96.3    | 99.6  | 93.9       |
| 60  | 0.6    | Gaussian   | Null   | 21.2    | 13.1  | 12.3    | 16.8  | 7.3        |
| 60  | 0.6    | Gaussian   | −3.5%  | 63.3    | 50.9  | 48.8    | 53.3  | 35.7       |
| 60  | 0.6    | Gaussian   | −7%    | 92.7    | 87.0  | 85.7    | 87.2  | 75.9       |
| 60  | 0.6    | Skewed     | Null   | 24.4    | 17.9  | 16.9    | 42.5  | 13.4       |
| 60  | 0.6    | Skewed     | −3.5%  | 64.4    | 54.2  | 52.3    | 80.7  | 46.7       |
| 60  | 0.6    | Skewed     | −7%    | 90.1    | 84.3  | 83.2    | 96.5  | 79.1       |
| 60  | 0.6    | $t_3$      | Null   | 21.6    | 13.6  | 12.5    | 17.7  | 7.4        |
| 60  | 0.6    | $t_3$      | −3.5%  | 66.9    | 56.5  | 54.2    | 66.6  | 43.3       |
| 60  | 0.6    | $t_3$      | −7%    | 92.9    | 88.4  | 87.5    | 95.5  | 83.1       |
| 120 | 0.0    | Gaussian   | Null   | 5.1     | 5.5   | 5.9     | 4.0   | 0.7        |
| 120 | 0.0    | Gaussian   | −3.5%  | 93.3    | 93.6  | 93.8    | 78.8  | 65.7       |
| 120 | 0.0    | Gaussian   | −7%    | 100.0   | 100.0 | 100.0   | 100.0 | 99.8       |
| 120 | 0.0    | Skewed     | Null   | 7.8     | 8.0   | 8.4     | 86.3  | 4.2        |
| 120 | 0.0    | Skewed     | −3.5%  | 90.3    | 90.5  | 90.8    | 100.0 | 83.8       |
| 120 | 0.0    | Skewed     | −7%    | 100.0   | 99.9  | 99.9    | 100.0 | 99.9       |
| 120 | 0.0    | $t_3$      | Null   | 4.9     | 5.2   | 5.6     | 4.2   | 0.6        |
| 120 | 0.0    | $t_3$      | −3.5%  | 93.5    | 93.6  | 93.9    | 98.8  | 87.1       |
| 120 | 0.0    | $t_3$      | −7%    | 99.7    | 99.8  | 99.8    | 100.0 | 99.6       |
| 120 | 0.3    | Gaussian   | Null   | 11.0    | 7.9   | 7.4     | 7.9   | 2.9        |
| 120 | 0.3    | Gaussian   | −3.5%  | 87.2    | 81.7  | 80.2    | 74.4  | 60.5       |
| 120 | 0.3    | Gaussian   | −7%    | 100.0   | 100.0 | 100.0   | 99.6  | 99.2       |
| 120 | 0.3    | Skewed     | Null   | 14.8    | 11.0  | 10.4    | 71.9  | 7.4        |
| 120 | 0.3    | Skewed     | −3.5%  | 84.6    | 79.2  | 77.9    | 99.7  | 71.6       |
| 120 | 0.3    | Skewed     | −7%    | 99.6    | 99.2  | 99.2    | 100.0 | 98.6       |
| 120 | 0.3    | $t_3$      | Null   | 12.0    | 7.9   | 7.4     | 8.8   | 2.3        |
| 120 | 0.3    | $t_3$      | −3.5%  | 88.8    | 84.5  | 83.4    | 94.2  | 75.5       |
| 120 | 0.3    | $t_3$      | −7%    | 99.7    | 99.4  | 99.4    | 100.0 | 99.1       |
| 120 | 0.6    | Gaussian   | Null   | 20.4    | 12.5  | 10.2    | 15.9  | 7.2        |
| 120 | 0.6    | Gaussian   | −3.5%  | 78.9    | 67.5  | 62.8    | 68.8  | 53.9       |
| 120 | 0.6    | Gaussian   | −7%    | 99.2    | 98.2  | 97.2    | 97.7  | 95.0       |
| 120 | 0.6    | Skewed     | Null   | 23.4    | 15.9  | 13.4    | 53.8  | 11.9       |
| 120 | 0.6    | Skewed     | −3.5%  | 77.5    | 67.9  | 63.2    | 94.3  | 60.0       |
| 120 | 0.6    | Skewed     | −7%    | 97.9    | 95.6  | 94.3    | 99.9  | 93.2       |
| 120 | 0.6    | $t_3$      | Null   | 21.9    | 13.5  | 10.9    | 17.3  | 7.2        |
| 120 | 0.6    | $t_3$      | −3.5%  | 81.0    | 71.7  | 67.3    | 83.3  | 62.1       |
| 120 | 0.6    | $t_3$      | −7%    | 98.7    | 97.4  | 96.7    | 99.8  | 96.0       |

## S11.4. Cross-record dependence stress test

Proposition 1 uses joint invariance under independent rotations of all monthly records. The product condition additionally specifies their joint behaviour across records. The stress test uses the 121 observed site coordinates, the five Gaussian graphs and six records from two simulated summers. The June, July and August record lengths were 30, 31 and 31 days in each summer.



For summer  $y$ , let

$$U_y(u) = \sum_{k=1}^3 \sqrt{w_k} \{A_{yk} \cos(2\pi k u) + B_{yk} \sin(2\pi k u)\}, \quad u \in [0, 1),$$

where the coefficients are independent standard normal variables and  $(w_1, w_2, w_3) = (1, 0.45, 0.20)/1.65$ . The same coefficients were used for all three records in a summer. For month  $m$  with length  $n_m$ , the centred spatial field at  $t = 0, \dots, n_m - 1$  was

$$\mathbf{z}_{ymt} = \lambda U_y(t/n_m) \mathbf{g} + \sqrt{1 - \lambda^2} \boldsymbol{\varepsilon}_{ymt}, \quad \lambda \in \{0, 0.3, 0.6, 0.8\}.$$

The vector  $\mathbf{g}$  is the centred, unit-mean-square pattern proportional to  $0.8x + y$  in projected coordinates. The month-specific Gaussian noise used the first 12 square-root-eigenvalue-weighted eigenmodes of the centred spatial covariance  $0.95 \exp(-d/450) + 0.05I$ , where  $d$  is in km. The retained basis was rescaled to mean site variance one. Its circular moving-average filter had weights  $(0.20, 0.50, 1, 0.50, 0.20)$ , normalised to unit Euclidean norm. Regional-mean series  $X_{ymt}$  were generated from an independent random-number stream with a circular filter proportional to  $(0.25, 0.60, 1, 0.60, 0.25)$ . The simulated field is  $\mathbf{y}_{ymt} = X_{ymt} \mathbf{1} + \mathbf{z}_{ymt}$ ; within-record type-7 quartiles of  $X_{ymt}$  define the high and middle labels.

Each individual record has a cyclically stationary Gaussian distribution. The shared Fourier coefficients align the three monthly phases within a summer, so independent month-wise rotations alter their cross-covariance when  $\lambda > 0$ . A numerical covariance check gave a maximum marginal one-step rotation error of  $2.0 \times 10^{-15}$ . Independently rotating the 30-day record by one day changed its cross-covariance with a 31-day record by 0.275 in relative Frobenius norm; the corresponding graph-dispersion cross-covariance changed by 0.362. The pooled same-summer July–August correlation of corresponding daily mean-dispersion values, averaged over the five graphs, increased from 0.003 at  $\lambda = 0$  to 0.769 at  $\lambda = 0.8$ .

For each value of  $\lambda$ , we generated 2,000 replications and used 999 independent product shifts per replication. The statistic averaged the record-specific high-to-middle relative contrasts (ratios minus one) equally over the six records and five graphs. Table S23 reports lower-tail rejection at  $\alpha = 0.05$ ; the seed was 20260812.

**Table S23.** Product-shift size under shared cross-record temporal dependence.

| Shared loading $\lambda$ | Rejection rate | Monte Carlo SE |
|--------------------------|----------------|----------------|
| 0.0                      | 0.0515         | 0.00494        |
| 0.3                      | 0.0510         | 0.00492        |
| 0.6                      | 0.0495         | 0.00485        |
| 0.8                      | 0.0490         | 0.00483        |

Rejection remained between 4.90% and 5.15% as the shared component increased. Thus rejection remained near its nominal level for these particular simulated departures from joint product invariance. This result does not verify the conditional product-invariance null for the observed climate record.

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